This material is neither intended to be distributed to Mainland China investors nor to provide securities investment consultancy services within the territory of Mainland China. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan.

Defu - A

From battery to AI: Initiate at OW with a positive view on copper foil

Initiation

Overweight

301511.SZ, 301511 CH
Price (15 Oct 26):Rmb78.17

Price Target (Dec-27):Rmb108.00

We initiate coverage of Defu Technology at OW with a Dec-27 PT of Rmb108. Defu is a leading Chinese electrolytic copper foil producer with a large LiB copper foil base and rising exposure to higher-margin electronic circuit foil. After a c.54% correction from its June peak (vs. MSCI China -5%), we see an attractive entry point ahead of a 3Q26E-2027E earnings growth cycle. We forecast NPAT to rise 5 times over 2026-2028, driven by PCB foil volume growth and HVLP mix upgrade, with carrier-attached foil upside and LiB foil as the earnings base. We believe the risk/reward is attractive with the stock at a 21x/14x FY27E/28E P/E.

- **Defu is well positioned in a tight HVLP4 market as the largest domestic producer.** AI server PCB/CCL upgrades require smoother HVLP copper foil to reduce high-speed signal loss, while HVLP4 is the highest-spec commercial grade currently used for next-generation AI server PCB/CCL layers. Qualified supply remains limited, given yield constraints, surface treatment capability and 18- to 36-month customer certification requirements. Defu passed HVLP3+ certification with Elite Material (EMC), Shengyi Technology (SYTech) and Panasonic in 1H26. We forecast customer-qualified HVLP4 supply shortages of ~0.7kt/month in 2026 and ~1.2kt/month in 2027.
- **PCB foil is the key driver, inflecting from 3Q26E and scaling through 2028E.** Defu is shifting from HTE-heavy products to higher-margin HVLP, driving our PCB foil profit forecast from Rmb0.2bn in 2026 to Rmb2.2bn in 2028. The ramp visibility is increasing, with ~700t HVLP shipments in September, including 180t+ HVLP4. With 50kt of new PCB foil capacity — RTF/HVLP — being added over 2027E-28E, we forecast a rise in PCB copper foil shipments from 25kt in 2026 to 64kt in 2028, and in HVLP share from <3% in 1H26 to 31%/51%/58% in 2H26/2027/2028. Carrier-attached foil adds 2027E upside, with c.65-70% GPM and potential net profit of c.Rmb157mn/ Rmb314mn in 2027E/28E, though contribution should be minimal in 2026.
- **LiB foil remains a material earnings base, with margin recovery extending into 2027E-28E.** Accounting for >80% of Defu's copper foil capacity, LiB foil benefits from stronger ESS demand, thinner-foil adoption and disciplined industry expansion. We forecast LiB foil GPM improvement from 8% in 2025 to 14% in 2028, with >90% utilization, supported by fees up c.Rmb2k/t YTD and price increase negotiations completed in late September with CATL, Defu's largest LiB foil customer.
- **Correction has reset expectations.** Defu's c.54% pullback reflects a market shift from AI exposure to earnings delivery and the broader AI hardware correction. While the HVLP ramp is increasingly recognized, we believe consensus still underestimates the speed and scale of Defu's high-end HVLP3+ commercialization. We expect 3Q26 earnings to rise 72% QoQ to Rmb200mn, with our 2026/27/28 estimates 0%/21%/5% above consensus. At 21x FY27E P/E, valuation does not fully capture the HVLP3+ ramp and carrier foil upside, in our view, while our SOTP-based Dec-27 PT of Rmb108 reflects stronger growth, technology barriers and AI infrastructure exposure.

Asia Basic Materials

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Sabrina Liu
(852) 2800-8535
sabrina.liu@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Frankie Fong
(852) 2800-8574
frankie.fong@jpmorgan.com
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

Technology

Jerry Tsai
(886-2) 2725-9867
jerry.tsai@jpmorgan.com
J.P. Morgan Securities (Taiwan) Limited

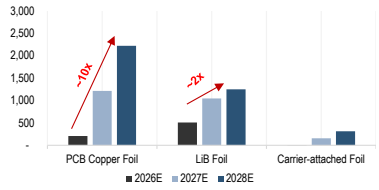
Style Exposure

Quant Factors	Current %Rank	Hist %Rank (1=Top) 6M	1Y	3Y	5Y
Value	84	58	58	94	98
Growth	30	100	100	1	75
Momentum	3	12	1		
Quality	81	94	94	14	
Low Vol	100	97	98	100	

LiB = Lithium-ion Battery
PCB = Printed Circuit Boards
CCL = Copper Clad Laminate
HVLP = Hyper Very Low Profile
HTE = High-temperature Elongation
RTF = Reverse Treated Foil

Defu - Segment net profit

Rmb mn



Source: Company data, J.P. Morgan estimates.

Sources for: Style Exposure – J.P. Morgan Global Markets Strategy; all other tables are company data and J.P. Morgan estimates.

See page 37 for analyst certification and important disclosures, including non-US analyst disclosures.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Price Performance



	YTD	1m	3m	12m
Abs	118.4%	-30.3%	-30.1%	147.5%
Rel	127.8%	-29.0%	-34.8%	163.2%

Company Data

Shares O/S (mn)	630
52-week range (Rmb)	178.88-27.86
Market cap ($ mn)	7,358
Exchange rate	6.70
Free float (%)	47.6%
3M ADV (mn)	33.10
3M ADV ($ mn)	452.2
Volatility (90 Day)	112
Index	SZBSHR
BBG ANR (Buy | Hold | Sell)	5|1|0

Key Metrics (FYE Dec)

Rmb in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	12,437	21,024	27,802	33,632
Adj. EBITDA	895	2,006	4,914	7,319
Adj. EBIT	491	1,586	4,424	6,711
Adj. net income	(32)	737	2,444	3,819
Adj. EPS	(0.05)	1.17	3.71	5.80
BBG EPS	0.17	1.17	3.20	5.80
Cashflow from operations	(381)	829	2,383	4,650
FCFF	(1,298)	(538)	283	3,450
Margins and Growth				
Revenue Growth Y/Y (%)	59.3%	69.0%	32.2%	21.0%
EBITDA margin	7.2%	9.5%	17.7%	21.8%
EBITDA Growth Y/Y (%)	363.8%	124.2%	145.0%	48.9%
EBIT margin	3.9%	7.5%	15.9%	20.0%
Net margin	(0.3%)	3.5%	8.8%	11.4%
Adj. EPS growth	(89.6%)	(2425.4%)	217.4%	56.2%
Ratios				
Adj. tax rate	22.8%	14.5%	15.0%	15.0%
Interest cover	4.4	7.6	24.1	44.9
Net debt/Equity	0.7	0.7	0.1	NM
Net debt/EBITDA	3.9	2.2	0.4	NM
ROCE	3.3%	9.5%	23.1%	30.7%
ROE	(0.8%)	16.6%	33.2%	32.7%
Valuation				
FCFF yield	(2.6%)	(1.1%)	0.6%	6.7%
Dividend yield	0.1%	0.1%	0.7%	1.1%
EV/Revenue	4.3	2.6	1.9	1.6
EV/EBITDA	60.4	27.6	10.9	7.2
Adj. P/E	NM	66.8	21.1	13.5

Summary Investment Thesis and Valuation

Investment Thesis

We initiate coverage of Defu Technology with an OW rating. Defu is a leading Chinese copper foil producer with a large LiB copper foil base and increasing exposure to higher-margin electronic circuit copper foil. After a meaningful share-price correction, we see an attractive entry point ahead of an earnings growth cycle. We expect growth to be driven by PCB copper foil mix upgrades and capacity expansion, carrier copper foil commercialization, and a recovery in LiB copper foil profitability. While the market remains focused on the pace of earnings realization, we think key leading indicators — downstream certification, volume ramp and product mix improvement — are moving in the right direction, supporting potential earnings acceleration and valuation re-rating.

Valuation

We derive our Dec-27 PT of Rmb108/share using a SOTP framework. We value PCB and carrier copper foil at 20x/25x FY28E P/E, lithium battery copper foil at 15x FY28E P/E, and other businesses at 10x P/E, reflecting differing growth prospects and profitability profiles across the portfolio. Our target price implies 29x/19x FY27E/FY28E P/E, between pure PCB copper foil and battery copper foil multiples, reflecting Defu's improving mix and earnings growth.

Source: J.P. Morgan Global Markets Strategy for Performance Drivers; company data, Bloomberg Finance L.P. and J.P. Morgan estimates for all other tables. Note: Price history may not be complete or exact.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Glossary

PCB	Printed Circuit Board	A PCB is the base board inside electronic products. Components are mounted on it, and the board connects them so that power and signals can flow.
CCL	Copper Clad Laminate	CCL is the base material used to make PCBs. It is an insulating board laminated with copper foil and later processed into circuit patterns.
Electronic Circuit Copper Foil	Electronic-circuit Copper Foil	Copper foil used in CCL and PCBs, later patterned into conductive circuits for servers, networking equipment, smartphones, automobiles and other electronics.
LiB Foil	Lithium-battery Copper Foil	Copper foil used as the current collector for the anode in lithium-ion batteries. It is a large-volume product category.
Carrier-attached Foil	Carrier-attached Copper Foil	A high-end copper foil with a detachable carrier that supports very fine circuit formation. It is used in optical modules, IC substrates, SLP and advanced HDI boards.
ED	Electrodeposition	ED is a process for making electrolytic copper foil by depositing copper from an electrolyte onto a rotating titanium drum using electric current.
RA	Rolling & Annealing	RA is another copper foil manufacturing method that uses rolling and heat treatment to produce thin copper foil.
Rz	Surface Roughness Rz	Rz is a measure of surface roughness. The report uses Rz to distinguish HTE, RTF and HVLP grades; a lower Rz generally means a smoother surface.
HTE	High-temperature Elongation	HTE is a more conventional electronic circuit copper foil with good elongation and reliability under high temperatures, commonly used in general-purpose PCBs.
RTF	Reverse Treated Foil	RTF is a specially treated copper foil designed to improve resin adhesion while supporting finer circuit processing.
HVLP	Hyper Very Low Profile	HVLP is a high-end copper foil with an ultra-smooth surface, used in products that require very high signal quality, such as AI servers and high-speed networking equipment.
HVLP1–5	Hyper Very Low Profile Grades 1–5	HVLP1 to HVLP5 are different grades of hyper-very-low-profile copper foil. Higher grades generally offer lower roughness and lower signal loss for more advanced high-speed applications.
HDI	High Density Interconnect	HDI is a denser and more advanced PCB technology that allows more circuits and connections in a smaller space. It is used in smartphones, automotive electronics and advanced consumer electronics.
SLP	Substrate-like PCB	SLP is a high-end PCB that is finer than a normal PCB and closer to an IC substrate in performance. It is often used in premium smartphones and compact electronics.
H-SLP / R-SLP / V-SLP	HTE / RTF / HVLP-based SLP Products	These refer to different SLP-related product series based on HTE, RTF or HVLP copper foil, used for applications such as BT substrates, SLP and HDI boards.
IC Substrate	Integrated Circuit Substrate	An IC substrate is the connection platform between a chip and the external circuit. It links the chip to the PCB and is an important material in advanced packaging.
BT Substrate	Bismaleimide Triazine Substrate	BT substrate is a special substrate material commonly used in chip packaging. It provides a stable connection base between the chip and external circuits.
mSAP	Modified Semi-Additive Process	mSAP is a PCB manufacturing process used to form finer circuit lines, commonly applied in premium smartphones, SLP and advanced packaging.
HPC	High Performance Computing	HPC uses powerful chips, servers and high-speed interconnects to process large and complex computing tasks, such as AI training and data-center workloads.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Company description:
Defu is a leading Chinese company that develops and manufactures copper foil used in printed circuit boards and lithium battery materials

The company is rapidly increasing its high-end PCB copper foil capacity to meet the accelerating demand growth from AI scaling

Investment summary

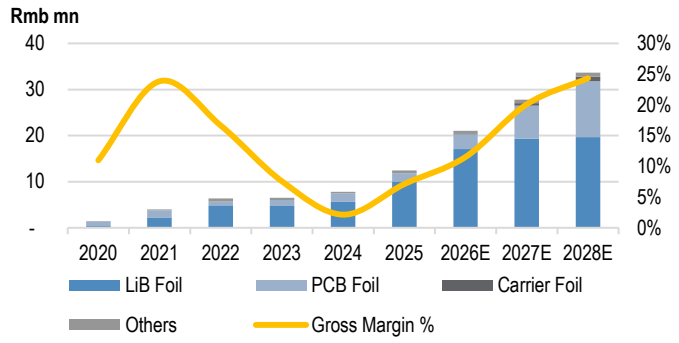
We initiate coverage of Defu Technology at OW with a Dec-27 PT of Rmb108. We like Defu for its improving earnings mix: The electronic circuit segment should benefit from rising exposure to higher-margin HVLP copper foil and the commercialization of carrier-attached copper foil, while the baseline LiB foil business should recover as industry supply-demand dynamics improve and processing fees rebound. We forecast NPAT to rise 5 times over 2026-2028.

Investment positives

- **PCB copper foil earnings to rise with mix upgrade and new capacity commissioning.** Defu is ahead of domestic peers in high-end PCB copper foil shipments, with batch shipments beginning in 3Q26. As of Sep'26, the company had shipped c.700t of HVLP copper foil and 180t+ of HVLP4 copper foil. Defu was also among the few domestic suppliers to pass HVLP3+ certification with key CCL manufacturers, including EMC, Shengyi and Panasonic, in 1H26. We forecast Defu's HVLP shipment share to rise from <3% in 1H26 to 31%/51%/58% in 2H26E/2027E/2028E, driving PCB copper foil's share of group gross profit from 8% to more than 50%.
- **LiB copper foil profitability bottoming.** Defu's LiB copper foil business remains a sizeable earnings base, supported by c.160kt of capacity (>80% of 2026 copper foil capacity) and improving industry utilization. Strong ESS demand, increasing adoption of thinner copper foil and more disciplined capacity growth across the industry should support a gradual rebound in processing fees. We forecast Defu's LiB copper foil gross profit margin to improve from 8% in 2025E to 14% in 2028E, providing a more stable earnings base alongside the faster-growing PCB copper foil business.
- **Carrier-attached copper foil to add a new high-margin growth driver.** Carrier-attached copper foil is a high-margin product used in IC substrates, substrate-like PCBs, advanced HDI boards and optical modules. Defu's carrier-attached copper foil business is still in the early stages, but commercialization from 2027E could provide an additional earnings driver.

Figure 1: Defu - Gross Profit and GPM Trend

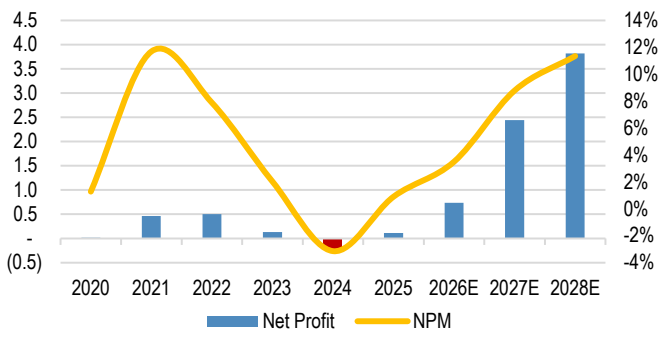
Rmb bn, %



Source: Company data, J.P. Morgan estimates.

Figure 2: Defu - Earnings and NPM Trend

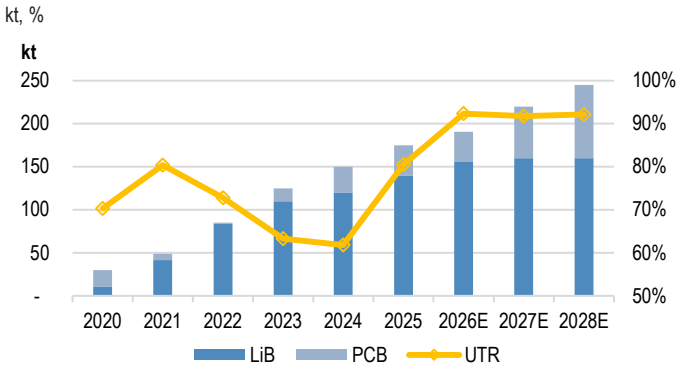
Rmb bn, %



Source: Company data, J.P. Morgan estimates.

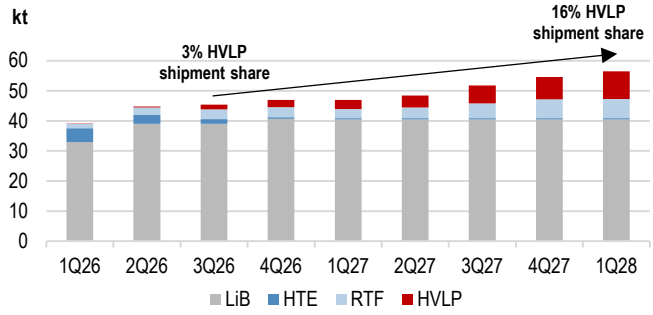
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 3: Defu - Copper Foil Capacity and UTR



Source: Company data, J.P. Morgan estimates.

Figure 4: Defu - Copper Foil Shipment Breakdown



Source: Company data, J.P. Morgan estimates.

3Q26 Earnings Preview

Defu released a positive profit alert on October 14, guiding for 9M26 NPAT of Rmb440–480mn, implying 3Q26 NPAT of Rmb177–217mn. We view the alert as broadly in line with market consensus and our expectations. The company is scheduled to release its full 3Q26 results on October 29, 2026. Our initial 3Q26 NPAT estimate is Rmb200mn, up 72% QoQ, supported by solid shipment growth and continued PCB foil mix upgrade. We forecast PCB foil and LiB foil shipments of 6.4kt and 39kt, respectively, with HVLP's share of PCB foil shipments rising from 4% in 2Q26 to 25% in 3Q26E. We estimate this mix upgrade will lift the average PCB foil processing fee from Rmb27k/t in 2Q26 to Rmb45k/t in 3Q26E, while we assume a normalized tax rate of 15% for the quarter.

Table 2: Defu - 3Q26 Earnings Preview

	1Q26	2Q26	3Q26E	4Q26E
Net Profit (Rmb mn)	147	116	200	272
QoQ	*221%*	*-21%*	*72%*	*37%*
YoY	*710%*	*467%*	*615%*	*493%*
Shipment (kt) - JPMe	**39.3**	**44.8**	**45.6**	**47.2**
PCB Foil	6.3	5.8	6.4	6.3
HVLP as % pf PCB Foil	*0%*	*4%*	*25%*	*38%*
LiB Foil	33.0	39.0	39.0	40.5
PCB Foil Processing Fee (k Rmb/t)- JPMe	**23**	**27**	**45**	**60**
QoQ		*19%*	*64%*	*34%*

Source: Company data, J.P. Morgan estimates.

Valuation, risks and key share price drivers

PT methodology

We derive our Dec-27 PT of Rmb108/share using a SOTP framework. We value PCB and carrier copper foil at 20x/25x FY28E P/E, lithium battery copper foil at 15x FY28E P/E, and other businesses at 10x P/E, reflecting differing growth prospects and profitability profiles across the portfolio.

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Key downside risks

Industry-wide risks include:1) slower-than-expected AI server deployment, 2) weaker demand for next-generation high-speed PCBs, 3) faster-than-expected high-end copper foil capacity additions, and 4) weaker LiB copper foil processing-fee recovery.

Company-specific risks include: 1) unexpected yield or qualification challenges during Defu's HVLP4 ramp, 2) delays in carrier copper foil commercialization, 3) slower utilization recovery in LiB copper foil, and 4) higher-than-expected balance-sheet pressure during the capacity expansion phase.

Key share price drivers and catalysts

Key catalysts are: i) HVLP qualification conversion into batch shipments and earnings, especially HVLP4; ii) ramp progress of the 50kt high-end electronic circuit copper foil project; iii) carrier copper foil order/shipment progress; iv) quarterly PCB copper foil processing-fee and product-mix trends; v) LiB copper foil utilization and processing-fee recovery; and vi) FCF inflection and deleveraging from 2027E.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Copper Foil Industry Primer

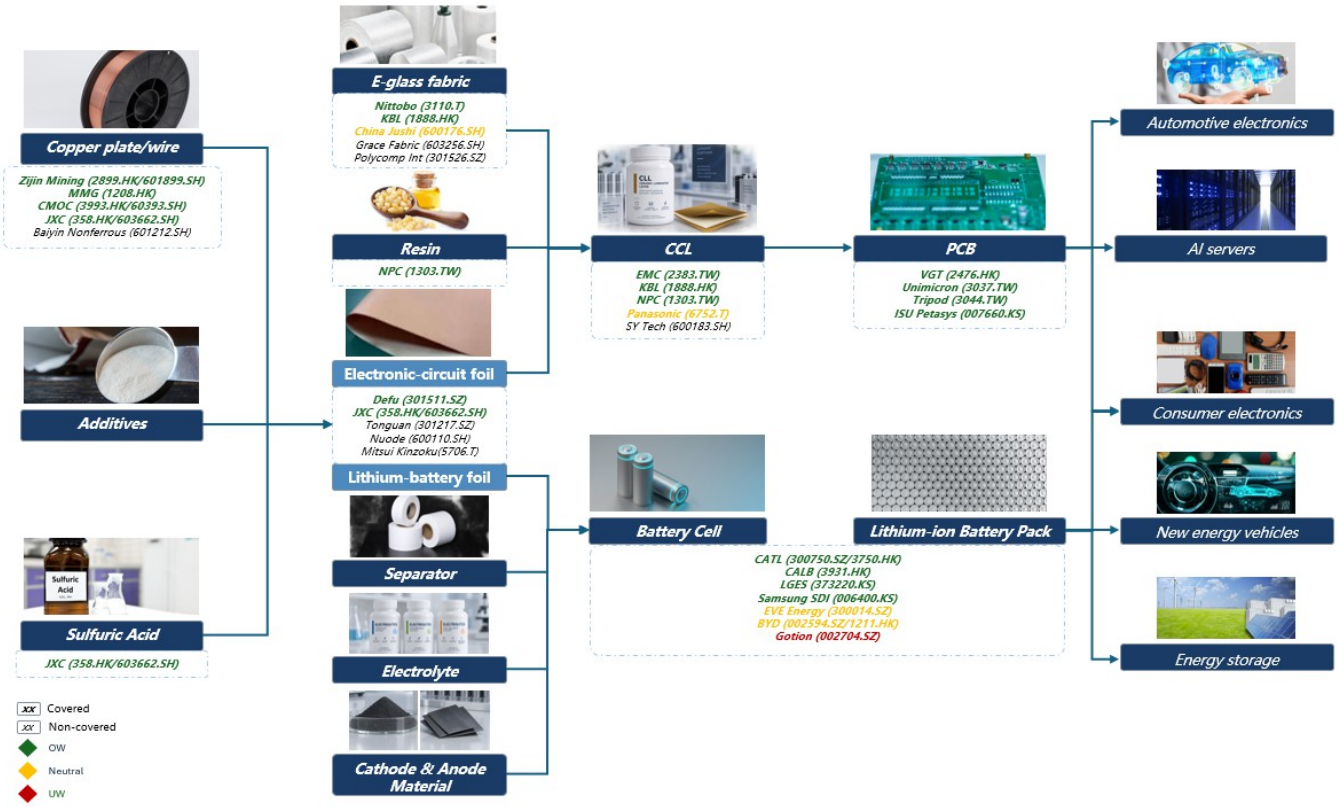
Printed circuit board (PCB) is the physical board that supports and connects electronic components. It enables signal transmission and power distribution, linking processors, memory, storage, and communication modules into a functioning electronic system.

Copper clad laminate (CCL) is the base material used to make PCBs. It consists of copper foil laminated onto an insulating substrate made mainly of resin and glass fiber cloth. Copper provides conductivity, resin provides insulation, and glass fiber provides mechanical strength.

What is copper foil?

Copper foil is a thin, high-purity, and continuous sheet of copper used as an electrical conductor. Its two principal applications are: 1) Electronic-circuit copper foil, which is laminated into copper-clad laminate (CCL) and subsequently patterned into the conductive circuits of a printed circuit board (PCB), and ultimately assembled into servers, networking equipment, smartphones, automobiles, and other electronic devices. 2) Lithium-battery copper foil, which serves as the current collector for the anode of a lithium-ion battery (LiB). Defu participates in both markets and is expanding into carrier-attached copper foil, a separate high-margin product category for fine-line circuit formation.

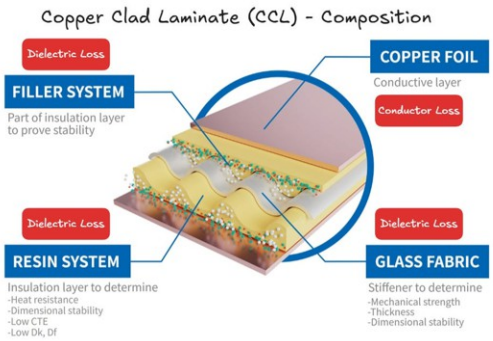
Figure 5: Industrial Chain of the Electrolytic Copper Foil Industry - JPM coverage



Source: Company reports, J.P. Morgan.

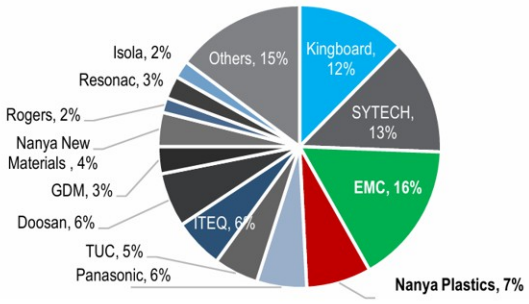
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 6: CCL Design



Source: Doosan Electronics

Figure 7: Global rigid CCL market share by revenue, 2025



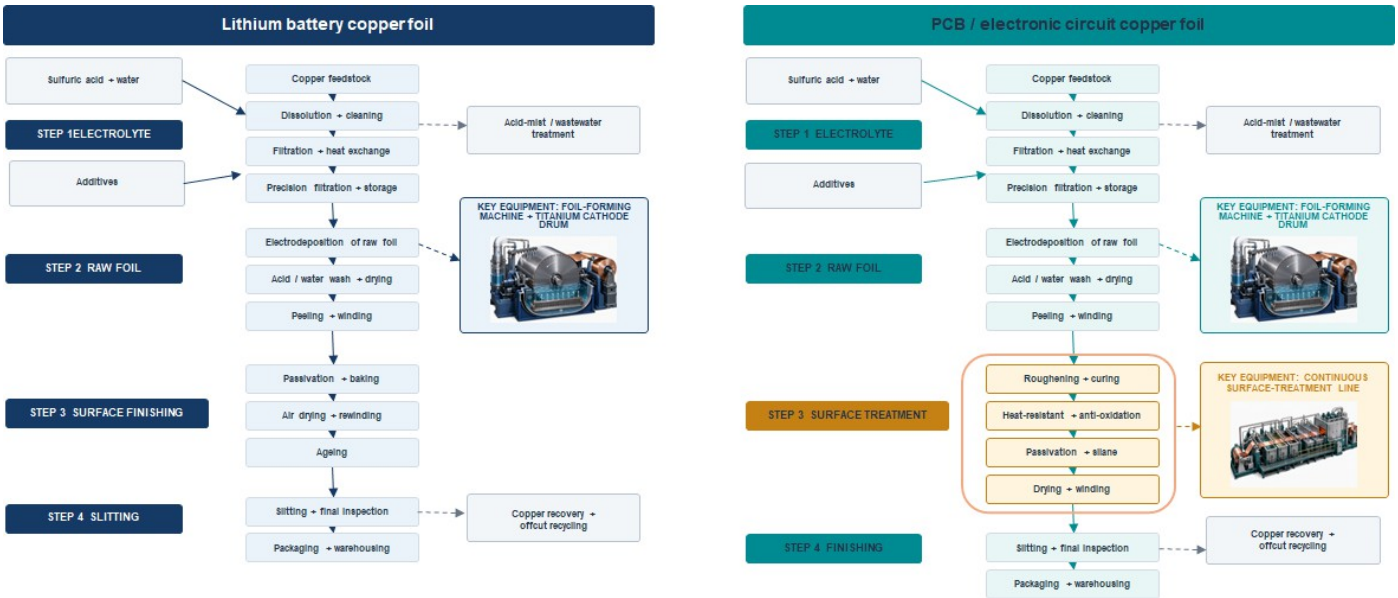
Source: Prismark (via Elite Materials Aug 2026 company presentation), J.P. Morgan

How is copper foil produced?

Copper foil is manufactured using two main methods: electrodeposition (ED) and rolling & annealing (RA). Copper foil produced through ED is known as electrolytic copper foil, and its production process has four main stages: 1) electrolyte preparation, 2) electro-deposition, 3) surface treatment, and 4) slitting and inspection. In the ED process, copper is dissolved in sulfuric acid and deionized water to form a copper sulfate electrolyte, then deposited onto a rotating titanium drum under direct current. Key process parameters, including current density, drum speed, additives and tension control, determine the foil's thickness, crystal structure and performance.

The main process difference between LiB copper foil and electronic circuit copper foil lies in surface treatment. LiB copper foil typically only requires anti-oxidation treatment, while electronic circuit copper foil also requires roughening and curing, in addition to anti-oxidation, to improve peel strength, chemical resistance and surface roughness.

Figure 8: Production Process: Lithium battery copper foil vs. PCB copper foil



Source: Company report, J.P. Morgan.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

What is HVLP copper foil?

HVLP copper foil (hyper-very-low-profile copper foil) is a specialized electrolytic copper foil with an ultra-smooth copper-dielectric interface, which helps reduce signal loss from the skin effect during high-frequency and high-speed data transmission. Successive generations (HVLP1 to HVLP5) offer progressively lower signal loss and are increasingly important for advanced interconnect applications such as AI servers, high-layer-count PCBs and high-speed networking equipment.

HVLP4 is currently the highest-specification HVLP copper foil in commercial production for next-generation AI server PCB/CCL layers. It commands processing fees of around Rmb160–200k/t (approximately 8–10x those of standard HTE 18μm copper foil).

Figure 9: Electronic Circuit Copper Foil Spec

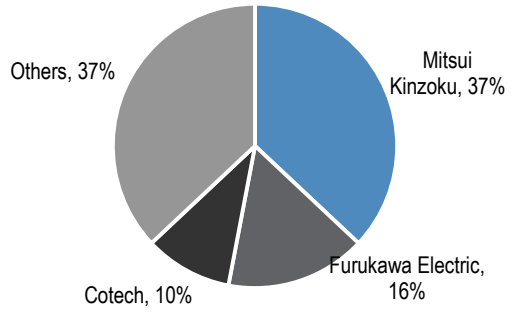
Foil Category	Grade	Surface Roughness (Rz)	Thickness	Processing Fee (Rmb/ton)	Typical Design Direction
High Temperature & Elongation (HTE)	HTE Foil	≥ 4.0 μm	18-70 μm	23,000	General-purpose PCB
Reverse Treated (RTF)	RTF 1	≤ 3.0 μm	12-35 μm	28,000	Mid- to high-speed PCB
	RTF 2	≤ 2.3 μm	12-35 μm	30,000	
	RTF 3	≤ 2.1 μm	12-35 μm	36,000	
	RTF 4	≤ 1.9 μm	12-35 μm	n.m.	
Hyper Very Low Profile (HVLP)	HVLP 1	≤ 2.0 μm	12-35 μm	70,000-80,000	Very-high-speed low-loss channels, 112G/224G, AI server, mmWave-class applications
	HVLP 2	≤ 1.5 μm	12-35 μm	80,000-100,000	
	HVLP 3	≤ 0.8 μm	12-35 μm	100,000-150,000	
	HVLP 4	≤ 0.6 μm	12-35 μm	160,000-200,000	
	HVLP 5	< 0.5 μm	12-35 μm	n.m.	
Carrier-attached		≤ 2 μm (Functional layer)	≤ 5.0 μm	ASP: Rmb80-120/sqm	IC Substrate (mSAP),1.6T optical module SLP boards, ultra-thin HDI for smartphones, memory/system semiconductors

Note: "Thickness" refers to commonly used copper foil thicknesses. Depending on supplier and application needs, HTE/RTF/HVLP copper foils may also include higher or lower thickness specifications for specialized high-frequency or thick-copper applications.
Source: SMM, Company data, J.P. Morgan

Qualified HVLP capacity is highly concentrated. The HVLP copper foil market is demonstrating a highly concentrated structure as the primary suppliers consist of a select few players, with Mitsui Kinzoku, Furukawa Electric and Co-Tech accounting for >60% of global HVLP supply in 2025. The manufacturing process for HVLP copper foil is highly complex because HVLP copper foil must meet the requirements for extremely low surface roughness, excellent adhesion, and high-frequency/high-speed signal integrity at the same time, which represent high technical barriers that prevent easy entry for newcomers. In 2026, Mainland Chinese producers such as Defu and Tongguan have also started to enter the AI supply chain with HVLP4 entering the batch shipment stage. After factoring in announced expansions and our assumed effective HVLP4 conversion rate, we estimate qualified suppliers can provide only ~0.8kt/month of HVLP4-equivalent supply by end-2026E and ~1.9kt/month in 2027E.

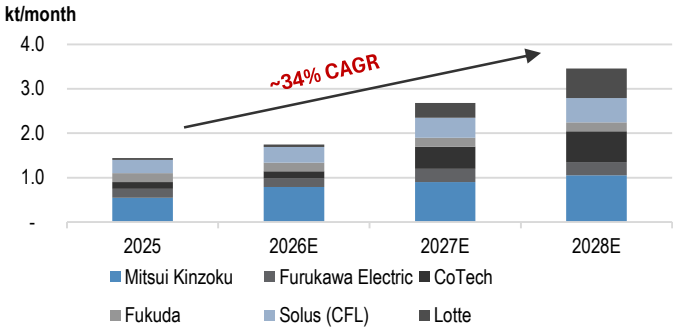
Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 10: HVLP CR3 market share (2025)



Source: Taiwan's Industrial Technology Research Institute (ITRI), J.P. Morgan

Figure 11: Major ex-mainland China suppliers' HVLP shipments



Source: Company data, J.P. Morgan estimates.

How does AI server evolution drive copper foil demand?

Electronic circuit copper foil is a critical upstream component in the PCB value chain, with its specifications increasingly dictated by AI server upgrade cycles. Copper-clad laminate (CCL) typically represents 40–60% of PCB raw material costs, while copper foil accounts for approximately 30–40% of the CCL cost, varying by laminate structure and thickness. As AI servers demand higher throughput, denser routing, and increased layer counts, signal-loss budgets have tightened significantly. At higher frequencies, copper surface roughness becomes a primary source of insertion loss, accelerating the migration from conventional HTE/RTF foils toward lower-roughness HVLP grades (HVLP1–5).

Table 3: Major AI server copper foil spec

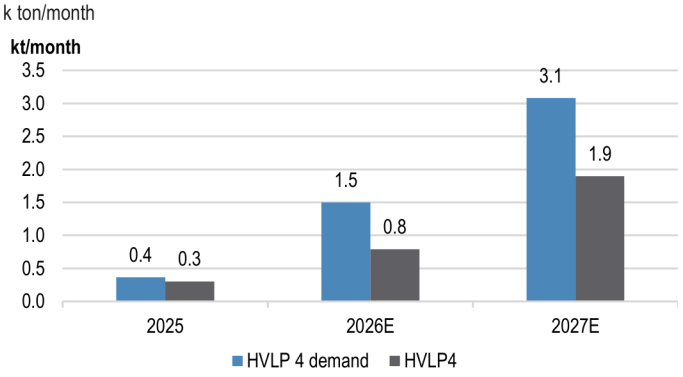
Platform	Ramp-up Year	Content	CCL	Copper foil
Nvdia				
H100/200	2022-2024	Compute Board	M6	HVLP2
		Switch Board	M6/7	HVLP2
GB200/300	2024-2026	Compute Board	M8	HVLP3
		Switch Board	M7/8	HVLP2/4
Vera Rubin	2H26E	Compute Board	M8	HVLP4
		Switch Board	M8.5	HVLP4
		Midplane	M9	HVLP4
Rubin Ultra	2H27E	Compute Board	M8.5/M9?	HVLP4
		Switch Board	M8.5/M9?	HVLP4
		Midplane	M9	HVLP4
		Backplane	M9/10/PTFE?	HVLP4/5?
Google				
v5/v5p	2023-2024		M7	HVLP2
v6/7	2H25		M7	HVLP2
v8	2H26E		M8.5/M8	HVLP4
v9	2028E		M9?	HVLP4/5?
Amazon				
Tranium 2	2H24-2026		M8	HVLP2
Tranium 3	2H26E		M8	HVLP4
Tranium 4	2028E		M9?	HVLP4/5?

Note: The above list is based on market feedback and expectations. Final design parameters and specifications may vary depending on company requirements and the chosen technology roadmap.
Source: Nvidia, Google, Amazon, J.P. Morgan estimates

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

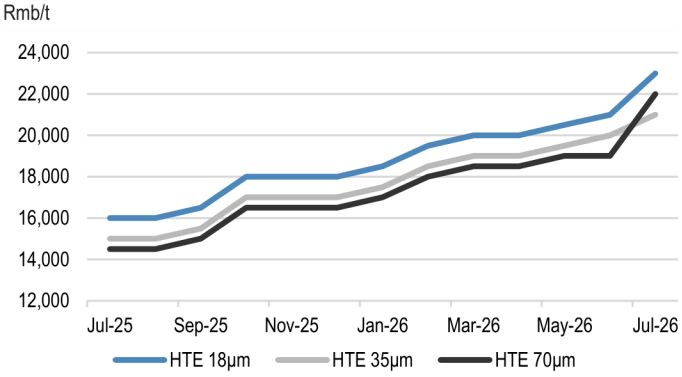
We anticipate significant HVLP4 shortages due to this transition: ~0.7kt/month in 2026E and ~1.2kt/month in 2027E. Our demand forecast is underpinned by AI server shipment growth (link to our Technology team's latest AI server shipment forecast), higher copper foil content per platform and a platform mix shift toward higher-grade HVLP foil. Industry feedback suggests NVIDIA's Vera Rubin platform will ramp in 4Q26E, establishing HVLP4 as the baseline specification for high-speed layers. Similarly, non-NVIDIA platforms—including AWS T3 and Google v8—are expected to adopt HVLP4 starting in 2H26E. Consequently, we forecast global AI-driven HVLP4 demand to rise from 0.4kt/month in 2025 to 3.1kt/month by 2027E.

Figure 12: HVLP4 Supply vs Demand



Note: i) The above includes only AI-driven demand and excludes demand from general servers; ii) Only certified HVLP4 suppliers are included.
Source: Company data, J.P. Morgan estimates.

Figure 13: Standard HTE Copper Foil Trend



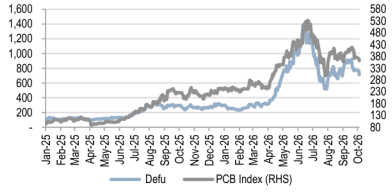
Source: SMM, J.P. Morgan.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Figure 14: Defu vs. China PCB index

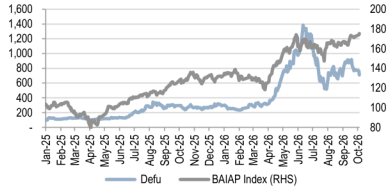
Stock/index price is rebased to Jan 1, 2025



Source: Bloomberg Finance L.P., Wind, J.P. Morgan.

Figure 15: Defu vs. Global AI Index

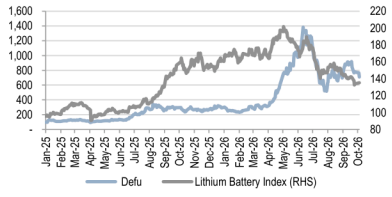
Stock/index price is rebased to Jan 1, 2025



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 16: Defu vs. China Lithium Battery Index

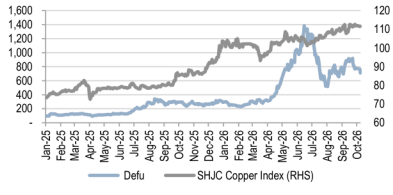
Stock/index price is rebased to Jan 1, 2025



Source: Bloomberg Finance L.P., Wind, J.P. Morgan.

Figure 17: Defu vs. Copper Price

Stock/index price is rebased to Jan 1, 2025
(Copper price unit: Rmb'000/t)



Source: Bloomberg Finance L.P., J.P. Morgan.

Share Price Analysis

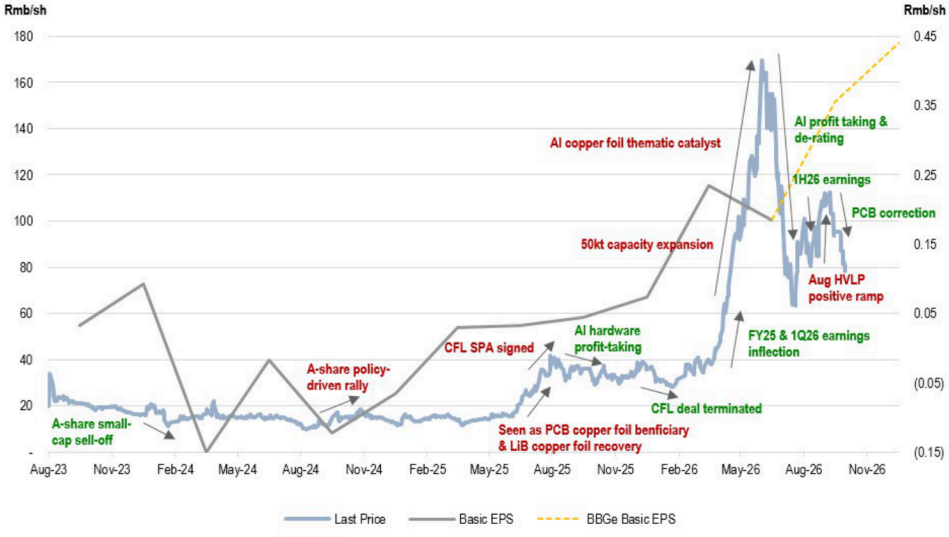
From industry beta to Defu's earnings delivery

The history. Defu's share price has seen two sharp rallies and two big corrections since its listing in 2023, reflecting changing investor expectations for the copper foil cycle. After falling c.65% from its IPO price and bottoming in early 2024, the stock traded through one of the industry's most challenging periods, as aggressive capacity additions, weak utilization and intense price competition pushed the sector into losses. Fundamentals began to recover from 2H25, driven by improving LiB copper foil demand, supported by rising EV penetration and ESS deployment, and rapid growth in high-end PCB copper foil demand driven by AI server upgrades.

The rally. The stock's strongest gains occurred between mid-2025 and 1H26. Initial optimism was driven by the proposed CFL acquisition, which was viewed as a potential route to overseas high-end copper foil technology and customers. This momentum was further amplified by the AI-driven PCB copper foil opportunity, as expectations for tighter HVLP supply, accelerating qualification progress and stronger shipment growth boosted investor interest. Defu's share price rose more than 12x from its February 2024 trough and more than 4x in 1H26, reaching a peak of c.Rmb170/share in June 2026.

The correction. Since the June 2026 peak, however, the stock has corrected c.54%, materially underperforming earlier expectations for AI-related materials beneficiaries. We attribute the pullback to both sector and company-specific factors. Sector-wide, investors became increasingly concerned about AI deployment timing, Rubin-related delays, near-term hardware digestion and cloud capex sustainability following the rapid AI materials rally. Company-specific concerns included the termination of the CFL acquisition, the proposed Rmb2.8bn placement and weaker-than-expected 2Q26 earnings. In our view, the market has shifted from rewarding AI exposure to demanding evidence of continuous earnings delivery.

Figure 18: Defu - Historical Share Price Trend with Key Events



Source: Bloomberg Finance L.P., J.P. Morgan

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Sep Site Visit Takeaway

We visited Defu Technology's Jiujiang Amber base on September 9, including a production facility tour and investor meeting with the Board Secretary and IR team. The visit mainly validated Defu's execution progress in high-end electronic circuit copper foil: HVLP shipments have reached measurable scale, HVLP4 yield is improving, self-developed surface-treatment equipment has entered normal production, and the 50kt high-end PCB copper foil project remains on track for phased start-up in 2Q27.

- **HVLP shipment ramp has moved beyond sampling.** August HVLP shipments reached c.500t, including c.70t of HVLP4. Management indicated that high-end electronic circuit copper foil capacity ramp-up has been stable and relatively fast. HVLP4 has entered an accelerated volume ramp-up phase and started to contribute commercial shipments. According to the latest company disclosure, Sep HVLP shipment reached c.700t (+40% MoM), including 180t+ of HVLP4.
- **HVLP4 yield is the key execution marker.** Defu's HVLP1–3 yield is close to 70%. HVLP4 yield is above 60% and still improving. Management attributed yield improvement to mass-production experience and process accumulation.
- **In-house equipment capability reduces a potential ramp bottleneck.** Defu currently uses both imported Mifune surface-treatment equipment and in-house developed equipment. Defu's self-developed surface-treatment equipment has entered regular production and benchmarks leading imported equipment on efficiency and finished-product yield. Management emphasized the equipment's process fit with Defu's own production system.
- **Single-surface treatment line output varies by thickness**. Management indicated that a surface-treatment line produces higher tonnage for 1oz products than for thinner products. HVLP4 1oz output can exceed 100t per month per line while HVLP4 0.5oz is c.40t per month per line.
- **The 50kt high-end PCB copper foil project is the next capacity milestone.** The new 50kt high-end electronic circuit copper foil project in Amber base is under construction and targeted for phased start-up in 2Q27.
- **Product mix upgrade is already visible in the operating focus.** Lower-end HTE products are being reduced, while RTF and HVLP products are the main PCB copper foil upgrade direction. Carrier copper foil remains in batch testing and customer introduction, with optical-module applications prioritized.
- **Customer feedback supports ramp visibility, but certification remains a gating factor.** HVLP shipment growth is being driven by selected leading CCL makers including EMC, SYTech and Panasonic. Management indicated that lithium copper foil lines are difficult to convert into HVLP3 or higher-grade PCB copper foil lines. Customer qualification requires approval from CCL / PCB customers and downstream terminal customers.
- **Pricing and hedging support earnings visibility.** Defu prices products under a "copper price + processing fee" model. HVLP products command higher processing fees than RTF and HTE, with HVLP4 at the premium end given higher technical difficulty and tighter supply. Defu maintains a raw-material hedging mechanism and retains limited copper price exposure, which keeps copper price fluctuations manageable for earnings.

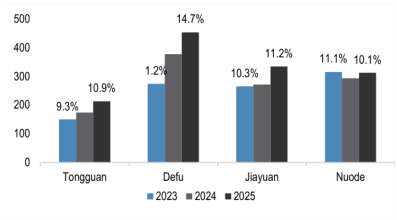
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Key Driver #1: PCB copper foil - HVLP shipment ramp and mix upgrade drive PCB foil earnings inflection

Defu is ahead of domestic peers in high-end HVLP copper foil, with the largest HVLP4 shipment scale among domestic suppliers. The company is among the few domestic suppliers moving beyond early-stage HVLP testing into high-end HVLP3+ certification and commercial ramp. As of 9M26, while domestic peers have made progress in HVLP1–2, HVLP3+ progress remains limited given much higher requirements for surface roughness, precision surface treatment, yield stability and customer qualification. By contrast, Defu has passed HVLP1–4 certification with EMC, Shengyi and Panasonic; in Sep, the company shipped c.700t of HVLP products (including c.180t+of HVLP4), with HVLP1–3 yield close to 70% and HVLP4 yield above 60% and still improving.

Figure 19: Defu vs Peers - R&D Employee %



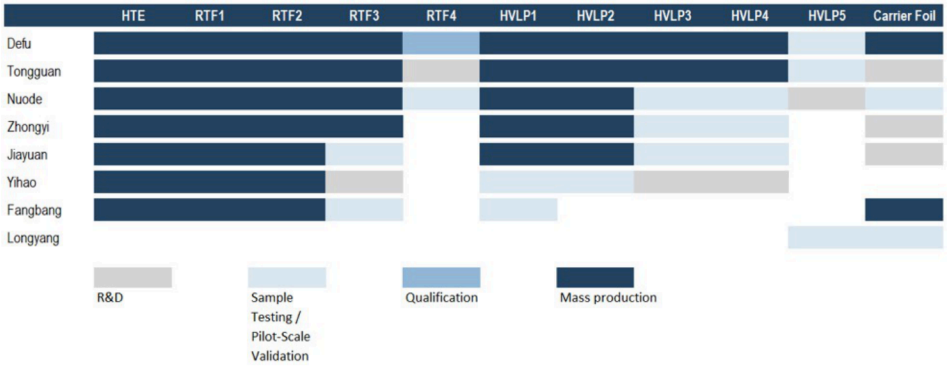
Source: Company reports, J.P. Morgan

Figure 20: Defu: Main PCB Copper Foil Products

	HTE	RTF	VLP	HVLP	Carrier	Embedded
IC Packaging/Assembly Products	H-SLP (BT, SLP, HDI)	R-SLP (BT, SLP, HDI)	V-SLP (BT, SLP, HDI)		C-IC1 (BT, SLP, HDI) C-IC1-C (SLP, Coreless)	
High Frequency Signal Products		R-HF1 (RTF1)		V-HF1 (HVLP1) V-HF2 (HVLP2)		
High Speed Transmission Products		R-HS1 (RTF1) R-HS2 (RTF2) R-HS3 (RTF3) R-HS4 (RTF3)		V-HS1 (HVLP1) V-HS2 (HVLP2) V-HS3 (HVLP3) V-HS4 (HVLP4)		
Flexible Circuit Boards		R-FPC				
Automotive Circuit Boards	H-HCD HTE-R (Thick)	R-HCD				
Electronics/Aerospace/Medical Products	H-HSP HTE-R (Thin)					E-TFR

Source: Company data, J.P. Morgan.

Figure 21: Product Progress of High-End Electronic Copper Foil by Leading Domestic Manufacturers



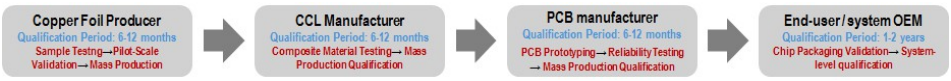
Note: The progress of non-covered companies is calculated based on company reports and disclosures.
Source: Company reports, J.P. Morgan calculations.

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Defu's advantage is underpinned by process know-how, vertical integration and sustained R&D investment. High-end HVLP requires tight control of copper crystal structure, additive formulation, surface roughness, adhesion and batch consistency. Defu has internalized key inputs through its subsidiaries Desi Optoelectronics (additives and chemical solutions), Stand Electrode Technology (consumables such as anode plates and separators), and Shuojin Energy (electro-deposition and surface-treatment equipment). We believe this integrated platform shortens Defu's process feedback loop and makes its high-end HVLP capability harder to replicate.

Surface-treatment capability and customer qualification reinforce Defu's lead in HVLP3+. Beyond process know-how, high-end HVLP capacity expansion requires access to advanced surface-treatment equipment and lengthy downstream qualification. HVLP3+ products require high-precision surface treatment and full qualification across CCL makers, PCB manufacturers and terminal customers (typically taking 18–36 months). In addition, high-end surface-treatment machines are harder to procure and scale than foil-forming machines, with external supply concentrated at Japan's Mifune (c.10 units of annual capacity and 24–30 month delivery lead times). Defu has secured key surface-treatment equipment with Mifune and developed in-house surface-treatment equipment, which our Sept. site visit indicates has entered stable production-line operation.

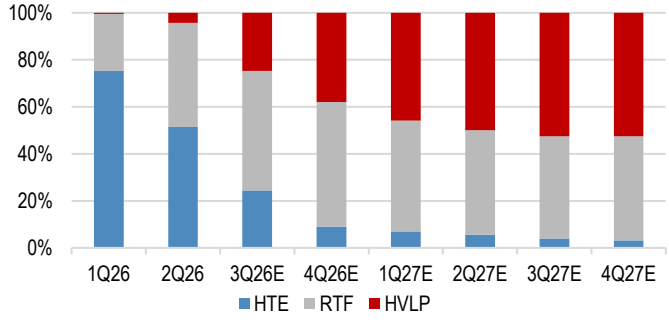
Figure 22: Qualification process



Source: J.P. Morgan

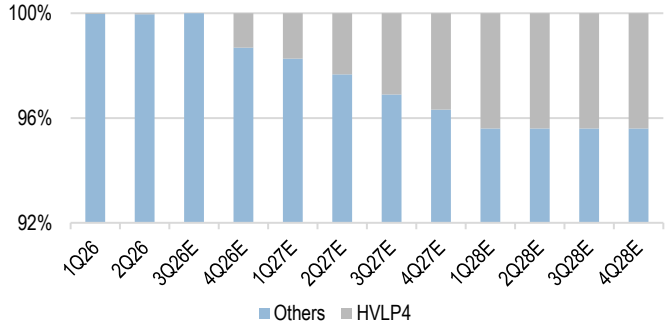
Qualification-to-shipment conversion to drive Defu's PCB copper foil earnings acceleration from 3Q26E. Against this backdrop, we forecast Defu's PCB copper foil earnings to accelerate from 3Q26E as qualification converts into shipments and the product mix shifts toward higher-margin HVLP. In 1H26, Defu's HVLP penetration was still low at <3%, with shipments mainly dominated by HTE and RTF. We expect the mix to begin improving in 3Q26E, and forecast HVLP shipment share to exceed 50% by 2Q27E, driving Defu's average PCB copper foil processing fee from Rmb39k/t in 2026E to Rmb86k/t in 2027E, and lifting PCB copper foil GPM from 21% to 38% over the same period.

Figure 23: Defu - PCB Copper Foil Product Mix



Source: Company data, J.P. Morgan estimates.

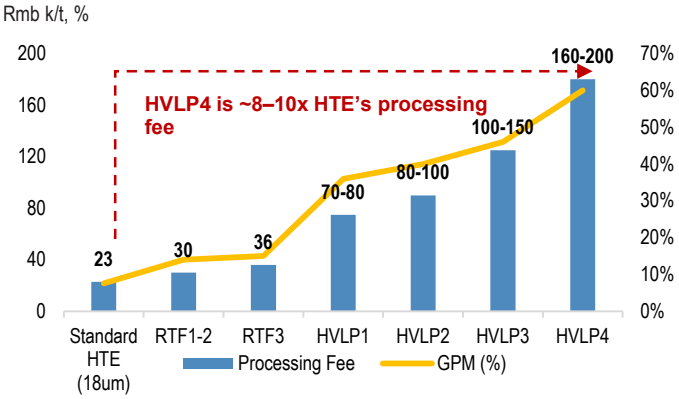
Figure 24: Defu - HVLP4 % As of Total PCB Foil shipment



Source: Company data, J.P. Morgan estimates.

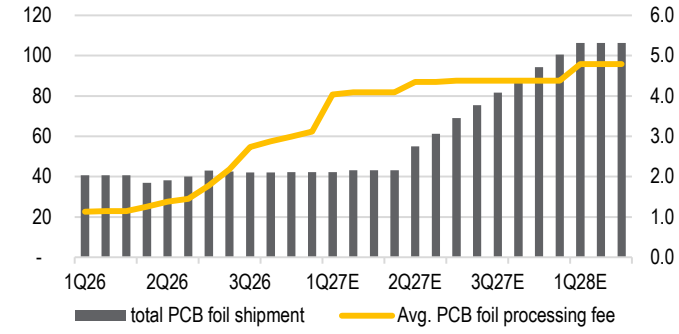
Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 25: Defu - PCB Foil Processing fee and GPM by product type (2026E)



Source: Company data, J.P. Morgan estimates.

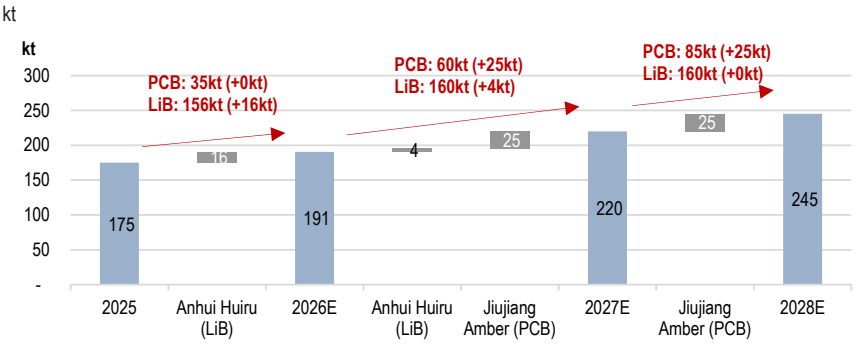
Figure 26: Defu - Monthly PCB copper foil processing fee trend (JPMe)



Source: Company data, J.P. Morgan estimates.

We expect capacity expansion to amplify Defu's HVLP mix upgrade and support a meaningful HVLP4 market share. Defu had 35kt of PCB copper foil nameplate capacity in 2026, and we forecast capacity to rise to 60kt/85kt in 2027E/2028E as its new 50kt high-end electronic copper foil project ramps up. We expect the project to use both imported and self-developed surface-treatment equipment, with self-developed equipments supporting the near-term HVLP4 ramp as newly ordered Mifune equipment is delivered and commissioned in 2H27. Based on our Sep site visit, each surface-treatment machine can support c.100t+/40t+ of monthly 1oz/0.5oz HVLP4 output, respectively. We estimate Defu could capture over 20% of global HVLP4 supply in 2027E, reflecting its climbing capacity and improving yield. After factoring in yield and conversion losses during the product-mix upgrade, we forecast total PCB copper foil shipments to rise from 24.6kt in 2026E to 63.8kt in 2028E, driving PCB copper foil gross profit from Rmb65mn in 2025 to Rmb2.8bn/Rmb4.8bn in 2027E/2028E.

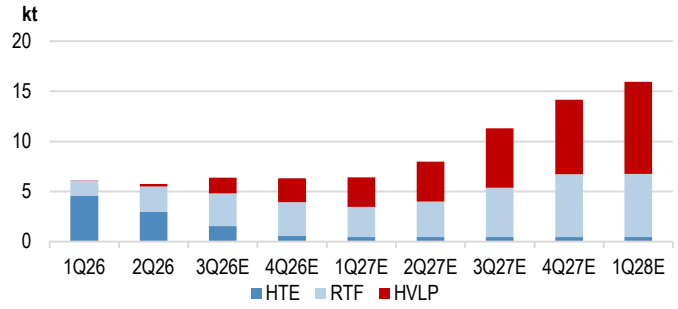
Figure 27: Defu: Copper Foil Capacity Expansion



Source: Company data, J.P. Morgan estimates.

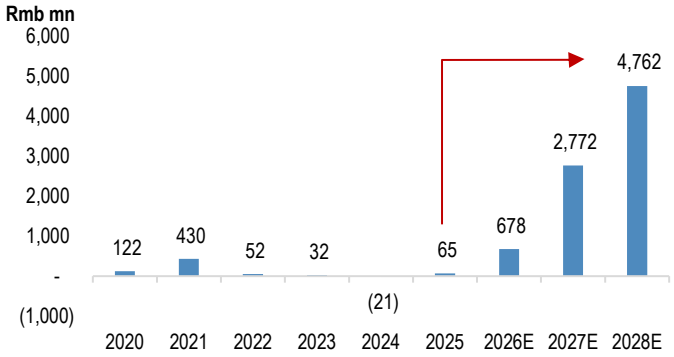
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 28: Defu - PCB Copper Foil Shipments



Source: Company data, J.P. Morgan estimates.

Figure 29: Defu - PCB Copper Foil Gross Profit



Source: Company data, J.P. Morgan estimates.

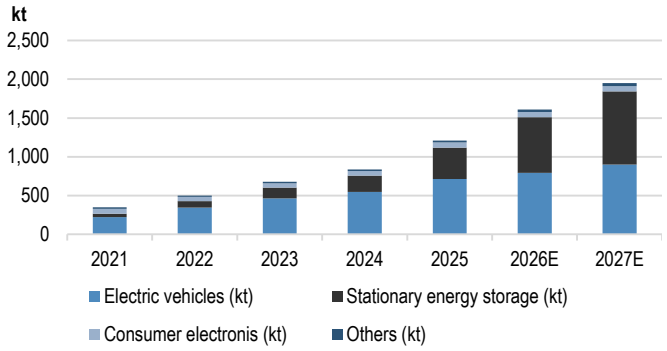
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Key Driver #2: LiB copper foil — recovering profitability provides a resilient earnings base

LiB copper foil remains Defu's largest volume base, and we expect a bottoming profit-repair cycle to emerge as demand improves and industry capacity growth becomes more disciplined. Defu's own LiB copper foil capacity is set to increase only modestly from 140kt in 2025 to 160kt in 2026E-2028E, implying limited incremental capacity additions. Despite this, we expect shipments to rise from ~119kt in 2025 to 152kt/162kt/162kt over the same period, mainly driven by higher utilization rates and technical upgrades. Unlike high-end PCB copper foil, LiB copper foil remains a more competitive product with stronger downstream battery-customer bargaining power, so we see this as a gradual profitability recovery rather than a sharp pricing cycle.

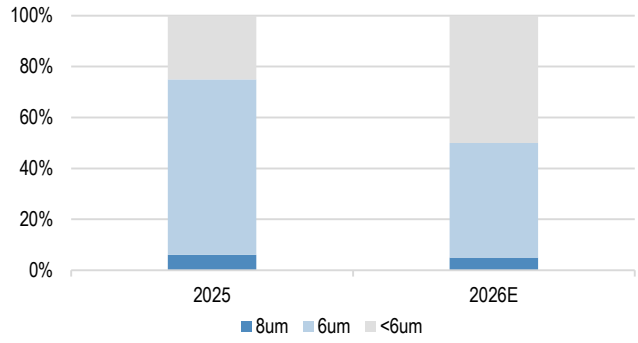
Demand remains resilient, led by ESS growth, overseas EV battery production and higher battery capacity per vehicle. Based on our Battery team's EV and ESS battery shipment forecast (link to the report), we estimate global LiB copper foil demand will rise from c.1.1mt in 2025 to c.1.5mt in 2026E and c.1.8mt in 2027E, implying 33%/21% y/y growth in 2026E/2027E. According to GGII, product mix is also improving as battery makers adopt thinner 5μm/4.5μm copper foil, which requires better process control and supports higher processing fees for qualified suppliers. According to the company, Defu has already started to benefit from this mix upgrade with its 5μm and 4.5μm shipments increasing in 2Q26, supporting LiB segment margin improvement.

Figure 30: Global LiB copper foil demand forecasts



Source: Company data, J.P. Morgan estimates.

Figure 31: GGII China LiB copper foil shipment forecast breakdown

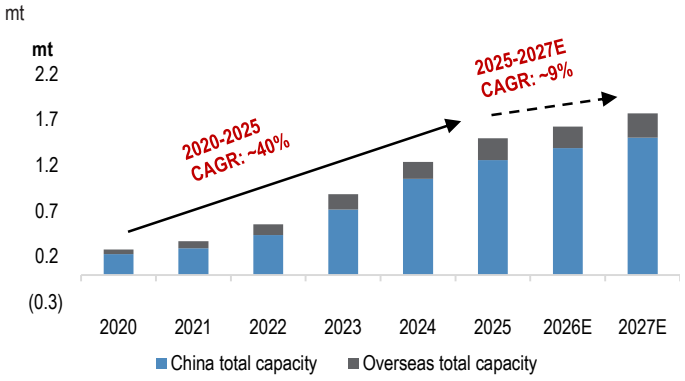


Source: Gaogong Industry Research Institute (GGII)

The supply side is also becoming more supportive. The previous LiB copper foil cycle was pressured by aggressive capacity additions, weak utilization and processing-fee compression. After several years of weak profitability, producers have become more cautious on capex, while some capacity is being reallocated toward higher-margin electronic circuit copper foil. We estimate industry capacity growth will be more disciplined over 2025–27E, helping utilization and pricing recover from trough levels.

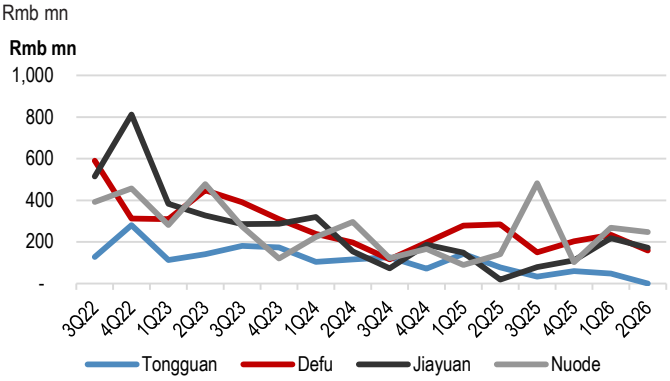
Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 32: 2020-2027E LiB copper foil capacity



Source: Company data, CCFA, J.P. Morgan estimates.

Figure 33: Copper Foil Company CAPEX trend

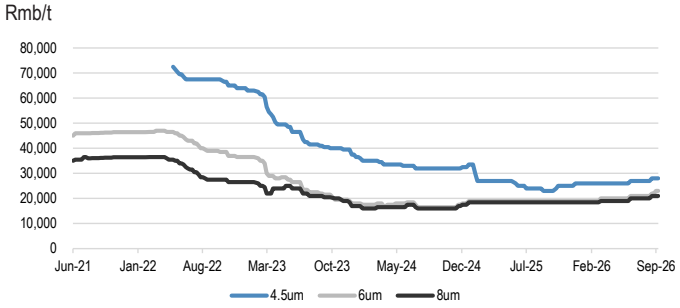


Source: Company data, Bloomberg Finance L.P., J.P. Morgan

We forecast LiB copper foil processing-fee improvement on two drivers: mix upgrade and price renegotiation amid a tighter supply-demand balance. On mix, higher adoption of 5μm/4.5μm foil should lift Defu's blended processing fee, as thinner products command higher fees than standard 6μm foil. On pricing, SMM data indicate LiB copper foil processing fees have risen by c.Rmb2k/t YTD across major thickness categories, while our supply chain checks suggest Defu completed Rmb1–2k/t price-increase negotiations with CATL in late September. We forecast Defu's average LiB copper foil processing fee to improve from Rmb15k/t in 2025 to Rmb20k/t/Rmb24k/t/Rmb26k/t in 2026E/2027E/2028E, driving unit GP from Rmb6.4k/t to Rmb11k/t/Rmb15k/t/Rmb16.5k/t.

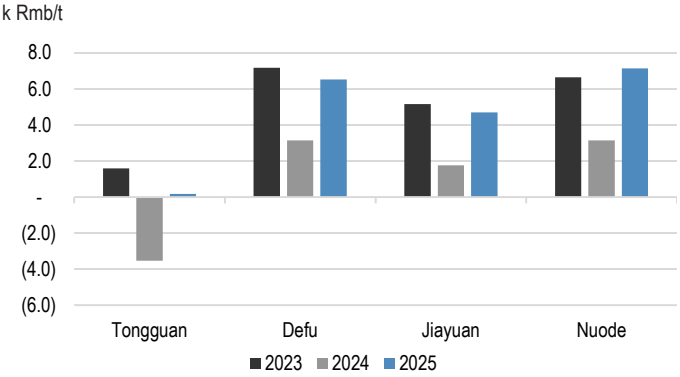
As a result, we estimate Defu's LiB copper foil gross profit will rise from Rmb766mn in 2025 to Rmb1.67bn/Rmb2.39bn/Rmb2.68bn in 2026E/2027E/2028E, with GPM improving from 8% to 10%/12%/14%. While the margin recovery is more gradual than in high-end PCB copper foil, LiB remains an important contributor given Defu's scale. We therefore see LiB copper foil as the earnings floor that supports Defu's transition toward a higher-margin PCB copper foil growth profile.

Figure 34: LiB Copper Foil Processing Fee



Source: SMM, J.P. Morgan

Figure 35: Defu vs Peers - Unit LiB Copper Foil GP



Note: Unit LiB copper foil GP are estimates based on company data
Source: Company data, J.P. Morgan estimates.

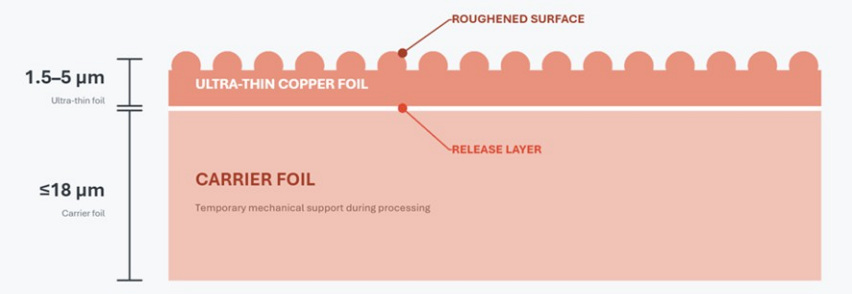
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Key Driver #3: Carrier-attached copper foil — a separate product category and potential medium-term profit driver

Carrier-attached copper foil is a separate high-end product from HVLP copper foil, used for fine-line circuit formation in optical modules, IC substrates, SLP and advanced HDI boards. It typically combines an ultra-thin copper layer, often ≤5μm, with a detachable carrier to support manufacturing stability. We see this new product as Defu's second high-margin growth option after HVLP, although contribution in 2026E should remain modest given the early commercialization stage. Defu's carrier foil is currently moving from repeated small-batch testing toward volume ramp, with optical modules the first target application, followed by substrate CCL and smartphone mSAP opportunities.

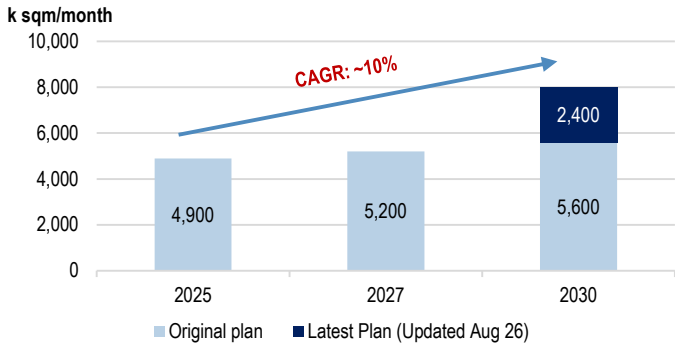
Carrier foil is a highly concentrated market, with Mitsui Kinzoku controlling over 90% of global capacity. Mitsui's plan to lift monthly MicroThin capacity from 5.6mn sqm to 8.0mn sqm by 2030 points to sustained demand growth, while applications are expanding from smartphone substrate-like PCBs into memory, optical modules and advanced packaging. Against this backdrop, Defu's entry represents an expansion into a new, growing, high-margin product market, with room for a qualified alternative supplier as downstream customers seek supply diversification.

Figure 36: Carrier-attached copper foil display



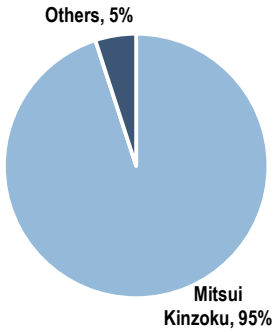
Source: Company data, J.P. Morgan.

Figure 37: Mitsui Kinzoku MicroThin product capacity



Note: Mitsui Kinzoku's MicroThin product is an ultra-thin functional copper layer ranging from 1.5 μm to 5 μm supported by a carrier foil, enabling high-density fine-circuit formation
Source: Mitsui Kinzoku, J.P. Morgan.

Figure 38: Carrier foil market share, 2025

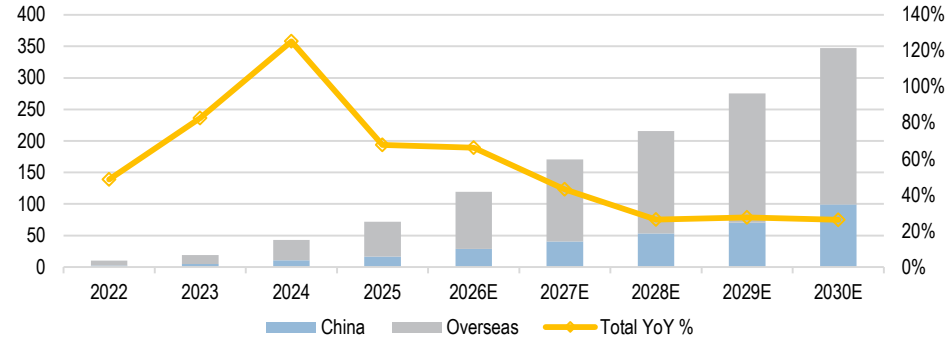


Source: Company data, J.P. Morgan.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 39: AI Optical Transceiver TAM, 2022-2030E

Rmb bn, %



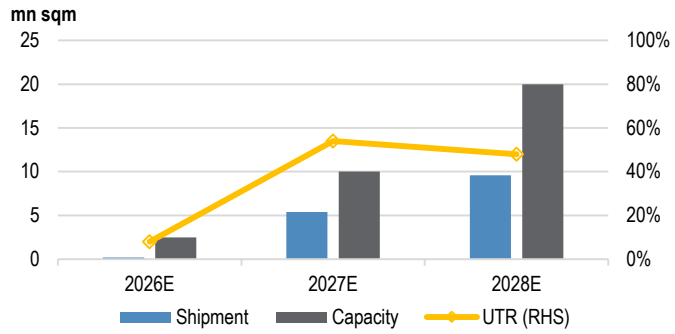
Source: Frost & Sullivan (via Crealights Technology 18 June 2026 Prospectus), J.P. Morgan.

Defu's carrier copper foil commercialization remains early. According to our site visit and Defu's latest announcement, its self-developed carrier copper foil product, C-IC2, has entered mass production for 1.6T optical-module projects. Management guided batch testing could be completed by 4Q26; however, larger-scale shipment still depends on customer qualification and yield stabilization. Shennan Circuit is one of the key early customers, and Defu is also working with leading optical-module and substrate customers on testing.

We expect Defu's carrier copper foil shipments of c.0.2mn sqm in 2H26E, rising to c.5.4mn sqm in 2027E and c.9.6mn sqm in 2028E. Based on a 65%-70% GPM assumption, we estimate carrier copper foil net profit contribution of c.Rmb4mn/ Rmb157mn/Rmb314mn in 2026E/2027E/2028E, respectively. The modest 2026E contribution mainly reflects a short commercial shipment window, while volume ramp-up and a full-year contribution support a more visible earnings step-up in 2027–28E.

Figure 40: Defu - Carrier Foil Capacity/Shipment/UTR

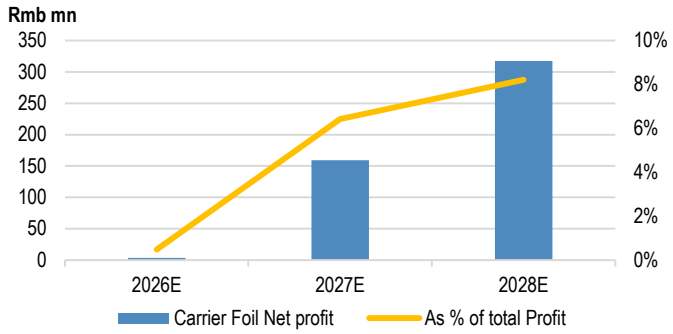
mn sqm, %



Note: Carrier-attached foil capacity is from Defu's company guidance
Source: Company data, J.P. Morgan estimates.

Figure 41: Defu - Carrier foil net profit contribution

Rmb mn, %



Source: J.P. Morgan estimates.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Valuation: SOTP to reflect diverging growth profiles across businesses

We use a sum-of-the-parts (SOTP) methodology as our primary valuation framework for Defu, reflecting the increasingly different growth and profitability profiles of its PCB copper foil, carrier copper foil and lithium battery copper foil businesses. Our Dec-27 target price of Rmb108 is based on FY28E SOTP values.

We value PCB copper foil and carrier copper foil at 20x/25x FY28E P/E, in line with global peers, reflecting their stronger earnings growth, higher technology barriers and increasing exposure to AI-related infrastructure demand. We value carrier-attached copper foil at a higher valuation in 2028 as we believe the segment will be the major driver of earnings growth in the medium term. We value lithium battery copper foil at 15x FY28E P/E, reflecting a cyclical earnings recovery but lower structural growth versus PCB copper foil. Other businesses are valued at 10x P/E. Our target price implies 29x/19x FY27E/FY28E P/E, between pure PCB copper foil and battery copper foil multiples, reflecting Defu's improving mix and earnings growth.

Table 4: Defu - SOTP Valuation

Breakdown	Implied Value (Rmb/sh)	% of target price	Valuation methodology
PCB Copper Foil	67.5	62%	20x FY28E segment P/E
Carrier Copper Foil	11.9	11%	25x FY28E segment P/E
Lithium Battery Copper Foil	28.5	26%	15x FY28E segment P/E
Other Business	0.5	0%	10x FY28E segment P/E
Target price	**108**	**100%**	

Source: Company data, J.P. Morgan estimates.

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Table 5: Global PCB value chain valuation comps

	Ticker	JPM Rating	Price Target	Upside/ downside	Share price	Perf. YTD (%)	Mkt Cap (USDm)	P/E (x) 2026E	P/E (x) 2027E	P/E (x) 2028E	P/B (x) 2026E	P/B (x) 2027E	P/B (x) 2028E	EV/EBITDA (x) 2026E	EV/EBITDA (x) 2027E	EV/EBITDA (x) 2028E
PCB Maker																
Ex-Mainland China																
IsuPetasys	007660 KS	OW	165000.00	71%	96300.00	-19%	900,913	25.6	16.1	12.5	6.9	4.9	3.6	17.5	11.1	9.0
Gold Circuit	2368 TT	NC	-	-	1190.00	73%	79,305	28.3	16.4	11.8	11.6	8.8	5.6	n.a.	n.a.	n.a.
Unimicron	3037 TT	OW	1225.00	14%	1075.00	389%	224,854	49.0	26.2	15.4	12.9	9.7	6.3	31.4	15.3	n.a.
Nan Ya PCB	8046 TT	NC	-	-	1345.00	458%	110,757	76.7	27.9	15.9	15.6	11.4	7.4	n.a.	n.a.	n.a.
Gold Circuit	2368 TT	NC	-	-	1190.00	73%	79,305	28.3	16.4	11.8	11.6	8.8	5.6	n.a.	n.a.	n.a.
Tripod Technology	3044 TT	OW	730.00	25%	586.00	85%	39,252	17.9	13.8	12.3	4.7	4.2	3.2	10.1	7.8	n.a.
Mainland China																
Victory Giant Technology	300476 CH	NC	-	-	186.00	-35%	30,381	22.1	11.9	8.6	4.6	3.5	2.7	17.8	9.5	7.0
Victory Giant Technology	2476 HK	OW	500.00	216%	158.00	-	25,808	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
WUS Printed Circuit	002463 CH	NC	-	-	116.18	59%	33,411	36.2	23.0	13.1	10.3	7.5	5.0	24.7	14.7	9.9
Shennan Circuits	002916 CH	NC	-	-	318.80	37%	32,452	39.1	25.3	18.1	10.0	7.3	5.5	27.1	18.5	14.0
Avary Holding	002938 CH	NC	-	-	75.26	49%	26,065	33.7	22.1	15.7	4.6	3.9	3.4	18.2	12.3	9.4
Delton Technology	001389 CH	NC	-	-	126.08	54%	8,907	26.6	14.6	9.7	8.8	5.7	3.7	18.5	10.9	7.5
Delton Technology	1989 HK	NC	-	-	81.90	-	5,786	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Shenzhen Kinwong Electronic	603228 CH	NC	-	-	90.21	23%	13,201	39.4	19.1	12.9	5.7	4.6	3.6	25.4	12.9	8.7
Suzhou Dongshan Precision Manufacturing	002384 CH	OW	30.00	-76%	124.20	47%	33,995	28.6	11.4	6.8	7.8	4.9	3.0	18.5	8.1	4.8
Market cap weighted average (Ex-Mainland China)								45.5	22.4	14.1	12.3	9.2	6.0	14.1	7.2	0.4
Market cap weighted average (Mainland China)								26.7	15.3	10.1	6.2	4.5	3.2	17.8	10.4	7.3
Market cap weighted average								***33.9***	***18.0***	***11.7***	***8.5***	***6.3***	***4.3***	***16.4***	***9.2***	***4.7***
CCL Maker																
Ex-Mainland China																
Resonac	4004 JP	NC	-	-	15025.00	133%	366,017	25.7	17.7	14.4	3.6	3.1	2.6	n.a.	n.a.	n.a.
Panasonic	6752 JP	N	4900.00	24%	3963.00	96%	1,239,752	18.7	15.2	12.9	1.7	1.5	1.4	9.4	8.4	7.8
Kingboard Laminates	1888 HK	OW	65.00	70%	38.28	192%	15,374	13.4	8.3	7.4	5.1	4.0	3.3	10.2	6.6	6.0
Elite Materials	2383 TT	OW	7200.00	41%	5120.00	211%	233,802	43.5	21.8	13.5	22.8	13.6	8.0	33.6	18.2	n.a.
Nan Ya Plastics	1303 TT	OW	415.00	32%	314.50	422%	317,866	21.9	14.6	10.7	5.3	4.5	3.8	34.2	16.1	13.9
Taiwan Union Technology	6274 TT	NC	-	-	1625.00	229%	62,015	42.5	19.9	11.4	16.2	11.0	7.3	n.a.	n.a.	n.a.
ITEQ	6213 TT	NC	-	-	799.00	604%	37,023	44.3	21.8	18.5	11.5	8.5	6.9	n.a.	n.a.	n.a.
TUC	6274 TT	NC	-	-	1625.00	229%	62,015	42.5	19.9	11.4	16.2	11.0	7.3	n.a.	n.a.	n.a.
Mainland China																
Shengyi Technology	600183 CH	NC	-	-	114.61	60%	41,601	38.8	25.6	18.3	12.6	9.4	6.5	28.1	18.8	13.9
Zhejiang Wazam New Materials	603186 CH	NC	-	-	190.00	283%	4,452	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Goldenmax International Group	002636 CH	NC	-	-	71.09	325%	7,734	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Changzhou Zhongying Science & Technology	300936 CH	NC	-	-	61.00	56%	685	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Market cap weighted average (Ex-Mainland China)								28.6	17.0	12.2	9.5	6.5	4.5	19.7	10.8	6.1
Market cap weighted average (Mainland China)								29.7	19.5	13.9	9.6	7.2	5.0	21.5	14.4	10.6
Market cap weighted average								***28.8***	***17.4***	***12.5***	***9.5***	***6.6***	***4.6***	***20.0***	***11.4***	***6.9***
Electronic Cloth																
Ex-Mainland China																
Nitto Boseki	3110 JT	OW	5500.00	89%	2916.00	43%	70,092	25.5	18.7	14.2	2.8	2.6	2.3	12.0	9.4	6.8
Asahi Kasei	3407 JT	OW	2100.00	34%	1567.00	13%	272,739	12.0	11.3	10.2	1.0	0.9	0.9	7.2	6.7	n.a.
Shin-Etsu Chemical	3063 JT	NC	-	-	1073.00	31%	1,724	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Taiwan Glass Industry	1802 TT	NC	-	-	62.70	81%	23,237	27.3	14.3	10.2	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Baotek	5340 TT	NC	-	-	90.70	30%	2,253	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Glotech	5475 TT	NC	-	-	193.00	101%	3,127	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Fulltech	1815 TT	NC	-	-	127.50	42%	9,988	28.4	17.5	n.a.	5.0	3.8	n.a.	n.a.	n.a.	n.a.
Mainland China																
China Jushi	600176 CH	N	11.00	-68%	34.32	101%	20,531	18.2	11.5	9.9	3.7	3.0	2.5	12.3	8.6	7.3
Sinoma Science & Tech	002080 CH	NC	-	-	42.50	17%	10,658	23.5	15.6	12.8	3.1	2.7	2.4	14.2	10.7	9.5
Grace Fabric Technology	603256 CH	NC	-	-	101.32	175%	13,696	80.9	43.1	27.8	29.5	18.2	11.7	56.4	34.0	21.8
Chongqing Polycomp Int.	301526 CH	NC	-	-	22.26	218%	12,544	50.6	30.4	22.8	8.8	7.3	6.0	n.a.	n.a.	n.a.
Changhai	300196 CH	NC	-	-	14.76	1%	902	16.5	13.4	11.3	1.2	1.2	1.1	8.9	7.6	6.7
Shandong Fiberglass	605006 CH	NC	-	-	12.75	75%	1,245	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Feilihua	300395 CH	NC	-	-	74.82	-25%	5,878	48.8	33.1	24.0	7.1	6.1	5.0	33.9	23.5	17.0
Kingboard Holdings	148 HK	NC	-	-	47.20	61%	6,747	5.8	5.2	4.7	0.7	0.7	0.6	n.a.	n.a.	n.a.
Market cap weighted average (Ex-Mainland China)								17.9	12.9	9.3	1.3	1.1	0.7	5.3	4.6	0.9
Market cap weighted average (Mainland China)								37.5	22.4	16.5	9.3	6.6	4.8	19.2	12.5	9.1
Market cap weighted average								***32.2***	***19.8***	***14.5***	***7.2***	***5.1***	***3.7***	***15.4***	***10.4***	***6.9***
Copper Foil																
Ex-Mainland China																
Mitsui Kinzoku	5706 JT	NC	-	-	2260.00	28%	165,379	14.2	9.8	11.6	2.8	1.3	2.4	n.a.	n.a.	n.a.
Furukawa	5801 JT	NC	-	-	3376.00	237%	304,036	20.9	16.5	13.0	4.5	3.6	2.8	n.a.	n.a.	n.a.
Lotte Energy Materials	020150 KS	NC	-	-	52100.00	70%	347,687	n.a.	135.0	32.5	1.7	1.7	1.6	n.a.	n.a.	n.a.
Solus Advanced Materials	336370 KS	NC	-	-	8060.00	10%	72,120	21.5	188.3	16.6	0.8	0.9	0.9	n.a.	n.a.	n.a.
Co-Tech	8358 TT	NC	-	-	461.50	67%	14,856	47.0	20.0	12.4	11.4	9.1	6.2	n.a.	n.a.	n.a.
Mainland China																
Tongguan	301217 CH	NC	-	-	89.25	160%	11,057	131.3	65.9	40.3	12.8	5.9	5.5	n.a.	n.a.	n.a.
Defu	301511 CH	OW	108.00	38%	78.17	118%	7,363	66.6	24.4	13.5	8.0	6.2	4.4	n.a.	n.a.	n.a.
Jiayuan	688388 CH	NC	-	-	26.00	-8%	2,564	22.6	13.7	10.2	1.9	2.0	1.8	n.a.	n.a.	n.a.
Nuode	600110 CH	NC	-	-	10.17	43%	2,660	44.8	17.7	12.8	2.9	2.5	2.1	n.a.	n.a.	n.a.
Market cap weighted average (Ex-Mainland China)								20.8	25.7	14.0	4.6	3.4	3.0	-	-	-
Market cap weighted average (Mainland China)								89.6	41.9	25.6	9.0	5.2	4.4	-	-	-
Market cap weighted average								***51.5***	***32.9***	***19.1***	***6.6***	***4.2***	***3.6***	***-***	***-***	***-***
Resin																
Ex-Mainland China																
Nan Ya Plastics	1303 TT	OW	415.00	32%	314.50	422%	317,866	21.9	14.6	10.7	5.3	4.5	3.8	34.2	16.1	13.9
Mainland China																
Sichuan EM Technology	601208 CH	NC	-	-	43.15	59%	6,514	56.0	32.0	21.2	6.6	5.7	4.9	n.a.	n.a.	n.a.
Epoxy Base Electronic Material	603002 CH	NC	-	-	16.26	120%	2,756	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Jinan Shengquan Group	605589 CH	NC	-	-	31.74	12%	4,014	21.0	16.3	13.4	2.3	2.2	2.0	13.9	11.5	9.7
Chongqing Polycomp International	301526 CH	NC	-	-	22.26	218%	12,544	50.6	30.4	22.8	8.8	7.3	6.0	n.a.	n.a.	n.a.
Market cap weighted average (Ex-Mainland China)								21.9	14.6	10.7	5.3	4.5	3.8	34.2	16.1	13.9
Market cap weighted average (Mainland China)								42.0	25.4	18.5	6.3	5.3	4.5	2.2	1.8	1.5
Market cap weighted average								***26.9***	***17.3***	***12.7***	***5.6***	***4.7***	***4.0***	***26.2***	***12.5***	***10.8***
Copper Foil/ Electronic Cloth Equipment																
Kunshan Dongwei Technology	688700 CH	NC	-	-	49.42	40%	2,204	44.1	25.2	16.8	7.1	5.8	4.6	31.9	18.6	12.3
Shaanxi Aerospace Power Hi-Tech	600343 CH	NC	-	-	17.07	-65%	1,628	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Saurer Intelligent Technology	600545 CH	NC	-	-	3.35	31%	895	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Zhejiang RIFA Precision Machinery	002520 CH	NC	-	-	5.46	-13%	612	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Zhejiang Taitan	003036 CH	NC	-	-	35.06	126%	1,135	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Market cap weighted average (Mainland China)								15.0	8.6	5.7	2.4	2.0	1.6	10.9	6.3	4.2
Market cap weighted average								***15.0***	***8.6***	***5.7***	***2.4***	***2.0***	***1.6***	***10.9***	***6.3***	***4.2***
Defu								**66.6**	**24.4**	**13.5**	**8.0**	**6.2**	**4.4**	**n.a.**	**n.a.**	**n.a.**

Source: Bloomberg Finance L.P., J.P. Morgan estimates. Note: Data as of Oct 15, 2026. NC means not covered by JPM. NC estimates are from BBG consensus.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Table 6: China EV value chain valuation comps

	Ticker	JPM Rating	Price Target	Upside/ downside	Share price	Perf. YTD (%)	Mkt Cap (USDm)	P/E (x) 2026E	P/E (x) 2027E	P/E (x) 2028E	P/B (x) 2026E	P/B (x) 2027E	P/B (x) 2028E	EV/EBITDA (x) 2026E	EV/EBITDA (x) 2027E	EV/EBITDA (x) 2028E
Lithium																
Ganfeng Lithium - A	002460 CH	OW	56.00	40%	40.01	-36%	12,536	11.3	10.5	9.8	1.6	1.4	1.3	9.2	10.7	11.3
Tianqi Lithium - A	002466 CH	N	45.00	22%	36.80	-34%	9,421	7.8	7.8	9.0	1.2	1.1	1.0	3.7	3.3	3.4
Ganfeng Lithium - H	1772 HK	OW	40.00	56%	25.60	-51%	6,840	6.1	5.6	5.1	0.9	0.8	0.7	9.2	10.7	11.3
Tianqi Lithium - H	9696 HK	N	25.00	2%	24.46	-52%	5,340	5.4	5.3	5.4	0.7	0.6	0.6	3.7	3.3	3.4
Yongxing Material	002756 CH	NC	-	-	35.88	-34%	2,891	9.8	8.4	6.5	1.4	1.2	1.1	4.7	4.1	3.2
Market cap weighted average								***7.9***	***7.5***	***7.3***	***1.1***	***1.0***	***0.9***	***6.6***	***7.2***	***7.5***
Nickel and Cobalt																
CMOC - A	603993 CH	OW	26.00	63%	15.99	-20%	51,123	10.4	9.6	8.7	3.2	2.6	2.1	5.3	5.0	4.5
CMOC - H	3993 HK	OW	25.00	76%	14.19	-26%	38,689	7.9	7.3	6.5	2.4	1.9	1.6	5.3	4.9	4.4
Huayou Cobalt	603799 CH	NC	-	-	32.18	-53%	9,106	8.2	6.3	5.4	1.1	1.0	0.9	8.3	6.9	6.1
Market cap weighted average								***9.1***	***8.3***	***7.4***	***2.6***	***2.2***	***1.8***	***5.5***	***5.1***	***4.6***
Cathode & Precursor																
GEM	002340 CH	NC	-	-	5.91	-29%	4,506	15.8	11.7	9.3	1.3	1.2	1.1	10.6	8.6	7.4
CNGR	300919 CH	OW	45.00	31%	34.31	-26%	5,355	12.7	10.5	9.0	1.4	1.2	1.1	9.0	8.4	8.1
Hunan Yuneng	301358 CH	N	70.00	48%	47.19	-27%	5,978	7.6	7.0	6.3	2.1	1.7	1.4	6.3	5.9	5.4
XTC New Energy	688778 CH	NC	-	-	38.68	-50%	2,917	18.3	14.8	12.4	1.9	1.8	1.6	15.0	12.6	11.6
Easpring	300073 CH	NC	-	-	38.20	-34%	3,107	18.6	14.3	11.6	1.3	1.2	1.1	13.2	10.0	8.1
Ronbay	688005 CH	UW	24.00	12%	21.47	-39%	2,293	39.8	22.8	17.6	2.0	1.8	1.7	17.8	13.2	11.4
Dynanonic	300769 CH	UW	35.00	-6%	37.22	-15%	1,558	12.4	9.6	8.0	1.9	1.6	1.4	9.2	8.2	6.9
Market cap weighted average								***15.9***	***11.9***	***9.8***	***1.7***	***1.5***	***1.3***	***10.6***	***9.0***	***8.0***
Electrolyte																
Tinci	002709 CH	NC	-	-	29.46	-36%	8,975	9.5	8.7	7.7	2.6	2.1	1.7	7.0	6.4	5.7
Capchem	300037 CH	NC	-	-	56.29	7%	6,342	19.1	15.5	12.7	3.4	2.9	2.5	15.3	12.3	10.1
Market cap weighted average								***13.5***	***11.5***	***9.8***	***2.9***	***2.4***	***2.0***	***10.4***	***8.9***	***7.5***
Anode																
Putailai	603659 CH	OW	35.00	78%	19.66	-28%	6,300	13.0	9.7	7.8	1.8	1.6	1.3	12.3	8.6	7.0
Ningbo Shanshan	600884 CH	NC	-	-	10.78	-20%	3,618	16.2	13.8	13.1	1.0	1.0	0.9	8.4	8.0	7.5
Market cap weighted average								***14.2***	***11.2***	***9.7***	***1.5***	***1.3***	***1.2***	***10.9***	***8.4***	***7.2***
Separators																
Yunnan Energy	002812 CH	N	60.00	24%	48.39	-15%	7,096	20.8	11.9	9.9	1.7	1.5	1.3	12.5	9.9	8.1
Senior Tech	300568 CH	NC	-	-	14.31	-6%	2,878	29.1	16.4	12.8	1.9	1.8	1.6	14.1	10.2	8.5
Market cap weighted average								***23.3***	***13.2***	***10.7***	***1.8***	***1.6***	***1.4***	***13.0***	***10.0***	***8.3***
Battery Equipment & Copper Foil																
Yinghe Tech	300457 CH	NC	-	-	19.72	-29%	1,901	19.1	15.9	14.1	1.7	1.6	1.4	11.2	9.1	8.8
Nuode	600110 CH	NC	-	-	10.17	43%	2,660	44.8	17.7	12.8	2.9	2.5	2.1	22.5	14.2	11.4
Defu	301511 CH	OW	108.00	38%	78.17	118%	7,363	66.6	24.4	13.5	8.0	6.2	4.4	35.6	19.3	10.0
Jiayuan	688388 CH	NC	-	-	26.00	-8%	2,564	22.6	13.7	10.2	1.9	2.0	1.8	13.8	9.6	7.6
Market cap weighted average								***48.6***	***20.2***	***12.9***	***5.2***	***4.2***	***3.1***	***26.2***	***15.3***	***9.7***
Battery Makers																
CATL	300750 CH	OW	520.00	87%	278.50	-23%	192,582	13.4	10.8	9.0	3.1	2.6	2.2	8.4	7.0	5.7
EVE Energy	300014 CH	N	60.00	28%	47.04	-28%	15,278	14.0	10.4	8.5	2.0	1.7	1.5	11.6	9.6	8.2
Gotion	002074 CH	UW	20.00	-29%	28.16	-28%	7,637	25.7	19.6	15.0	1.7	1.5	1.4	18.8	16.8	15.2
CALB	3931 HK	OW	40.00	162%	15.28	-39%	4,047	8.7	5.7	4.5	0.6	0.5	0.5	12.6	10.0	11.7
Market cap weighted average								***13.8***	***11.0***	***9.1***	***3.0***	***2.5***	***2.1***	***9.0***	***7.6***	***6.3***
OEMs																
BYD - A	002594 CH	N	88.00	7%	82.24	-16%	112,050	19.2	14.6	12.2	2.7	2.3	2.0	5.8	5.5	5.0
BYD - H	1211 HK	N	88.00	21%	72.95	-23%	99,392	14.6	11.3	9.4	2.0	1.7	1.5	5.8	5.5	5.0
Li Auto (Class A)	2015 HK	UW	35.00	-21%	44.44	-31%	13,759	n.a.	34.7	19.7	1.7	1.5	1.6	n.a.	1.5	0.8
Li Auto (ADR)	LI US	UW	9.00	-22%	11.51	-32%	1,781	n.a.	35.3	20.0	1.7	1.5	1.6	n.a.	1.8	1.0
Xpeng (Class A)	9868 HK	N	45.00	21%	37.20	-53%	10,668	n.a.	n.m.	23.5	2.2	2.2	2.0	n.a.	n.a.	69.7
Xpeng (ADR)	XPEV US	N	11.50	20%	9.60	-53%	1,376	n.a.	n.m.	23.8	2.2	2.3	2.1	n.a.	n.a.	70.5
Leapmotor	9863 HK	N	36.00	3%	35.00	-28%	7,069	15.1	8.2	5.9	2.4	1.9	1.4	5.2	3.8	3.0
NIO	NIO US	N	4.50	30%	3.47	-32%	1,300	n.a.	33.6	14.8	7.6	7.0	4.7	n.a.	191.9	10.2
Market cap weighted average								***13.5***	***14.6***	***12.6***	***2.5***	***2.2***	***1.9***	***4.6***	***11.0***	***9.4***
Defu								66.6	24.4	13.5	8.0	6.2	4.4	n.a.	n.a.	n.a.

Source: Bloomberg Finance L.P., J.P. Morgan estimates. Note: Data as of Oct 15, 2026. NC means not covered by JPM. NC estimates are from BBG consensus.

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Key risks to our positive view

- **High-end PCB copper foil ramp could be slower than expected.** Defu's earnings acceleration depends on converting customer qualification into repeatable commercial shipments, especially for HVLP4. Slower customer allocation, yield improvement, equipment performance, surface-treatment capacity ramp or AI server platform deployment could delay the product-mix upgrade and processing-fee expansion.
- **Domestic competitors could qualify faster than expected in high-end PCB copper foil**. Our positive view assumes HVLP4 qualification remains difficult for most domestic peers in the near term. Faster-than-expected CCL/PCB customer qualification, yield improvement or surface-treatment equipment access by domestic competitors could accelerate new qualified supply, shorten Defu's lead time and pressure HVLP processing fees and unit GP.
- **Carrier copper foil commercialization could be delayed.** Carrier copper foil remains an earlier-stage product for Defu and requires stable ultra-thin copper control, release performance and process reliability. Delays in optical-module or memory qualification, slower customer ramp, or lower-than-expected production stability could reduce contribution from this potential second high-margin profit stream.
- **LiB copper foil profitability recovery could disappoint.** Weaker ESS demand, faster industry capacity additions, slower thinner-foil adoption or sustained downstream pricing pressure could weigh on LiB processing fees and unit profit recovery. Given LiB copper foil remains Defu's largest volume base, even modest changes in unit GP could have a meaningful earnings impact.
- **AI/PCB sector derating risk.** Defu's valuation is increasingly linked to AI hardware and PCB copper foil sentiment. A broader pullback in AI infrastructure, PCB/CCL or high-end copper foil names could lead to multiple compression, even if Defu's earnings ramp remains on track.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Table 7: Defu - Scenario Analysis

		Bull	Base	Bear
Valuation summary				
Fair Value (based on FY28E SOTP)		143	108	67
FV Implied FY27E P/E (x)		29x	29x	23x
FV Implied FY28E P/E (x)		19x	19x	17x
Earnings				
FY27E	NPAT (Rmb bn)	3.3	2.4	1.9
FY28E	NPAT (Rmb bn)	5.0	3.8	2.6
Shipment				
FY27E	**Total (kt)**	**211**	**207**	**200**
	LiB foil (kt)	162	162	162
	HTE (kt)	2	2	2
	RTF (kt)	18	18	18
	HVLP (kt)	22	20	19
	Carrier foil (mn sqm)	8	5	0
FY28E	**Total (kt)**	**243**	**235**	**228**
	LiB foil (kt)	162	162	162
	HTE (kt)	2	2	2
	RTF (kt)	25	25	25
	HVLP (kt)	39	37	34
	Carrier foil (mn sqm)	14	10	5
Unit GP				
FY27E	LiB foil (Rmb k/t)	19	15	13
	HTE (Rmb k/t)	19	14	13
	RTF (Rmb k/t)	24	18	17
	HVLP (Rmb k/t)	140	119	115
	Carrier foil (Rmb/sqm)	73	67	0
FY28E	LiB foil (Rmb k/t)	19	17	12
	HTE (Rmb k/t)	2	2	2
	RTF (Rmb k/t)	25	25	25
	HVLP (Rmb k/t)	39	37	34
	Carrier foil (Rmb/sqm)	77	70	70

Note: Bull case assumption: Faster HVLP ramp, stronger processing-fee expansion and higher carrier copper foil adoption. Bear case: Slower HVLP commercialization, weaker LiB profitability recovery, delayed carrier copper foil contribution and derating in all segments.
Source: Company data, J.P. Morgan estimates.

- **Balance sheet, financing and dilution risk remain relevant.** Defu remains capex- and working-capital intensive as it expands high-end PCB copper foil capacity, and any delay in earnings or FCF improvement could keep leverage elevated for longer. The company has announced a proposed Rmb2.8bn equity placement to fund its 50kt AI high-end electronic circuit copper foil project and working-capital needs. At an illustrative Rmb78/share, the placement would imply c.36mn new shares, or c.5.4% dilution on a post-issue share count basis, based on Defu's current 630.3mn share base. While the placement could support expansion and balance-sheet repair, it remains subject to approvals and may dilute existing shareholders.

Table 8: Defu - Proposed Equity Placement Details

Detail	
Proposed equity placement size	Up to Rmb2.8bn
Use of proceeds — AI copper foil project	Rmb1.98bn for the 50kt AI high-end electronic circuit copper foil project
Use of proceeds — working capital	Rmb0.82bn for working-capital replenishment
Regulatory cap	New shares capped at 30% of pre-issue share capital, or up to 188.3mn shares
Approval status	Subject to shareholder approval, Shenzhen Stock Exchange review and CSRC registration
Implied dilution using latest share price	
Current share outstanding (mn shares)	629
Current share price (Rmb/sh)	78.17
Implied new shares (mn shares)	36
New shares as % of current share base	5.7%
Post-issue dilution	5.4%

Note: Priced as of Oct 15th, 2026
Source: Company reports, J.P. Morgan estimates.

Related research

China Battery & Materials: Copper foil supply tightness resolved post price hike; October production outlook (28 September 2026)

ESS Demand: Against Consensus: Why we remain constructive on ESS into 2030 (19 September 2026)

TSMC: CoWoS and Advanced Packaging Updates (17 September 2026)

PCB, CCL, and Substrate: CPO concerns likely overdone (17 September 2026)

Asia Tech Strategy At a crossroads in the AI Infra trade (16 September 2026)

Global EV & Battery Update: August 2026: EV/PHEV mix rising, led by EU; ESS demand remains solid (14 September 2026)

Taiwan ODMs AI and general servers beat, PCs miss (11 September 2026)

Nan Ya Plastics Corp: 3Q electronic materials on track for >20% q/q growth, expecting pricing momentum to accelerate into 4Q (10 September 2026)

Nan Ya Plastics Corp: Share price overreaction to media comments about NYPCB halting BT substrate price hikes from 4Q26 presents a buying opportunity (3 September 2026)

China AI Infra Ecosystem: Unprecedented growth; lingering constraints (2 September 2026)

Nan Ya Plastics Corp: Surprise, NPC has PTFE CCL: Raise PT to NT$350 as tailwinds from copper foil materialize; >40% ROE by 2027E (1 September 2026)

Taiwan PCB, CCL, and Substrate: Change in AI HDI leadership, for now (13 July 2026)

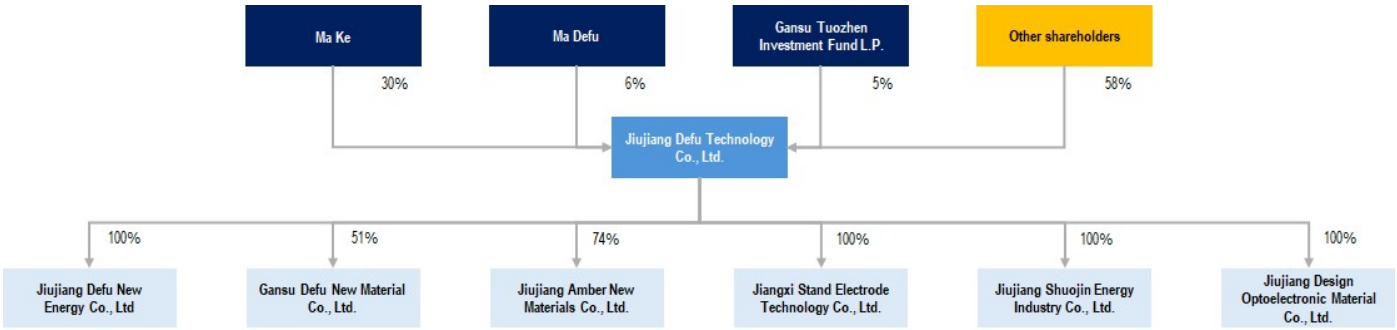
Elite Material Co: Entering the new era (22 May 2026)

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Company Background

Defu Technology is a leading Chinese copper foil producer, focused on high-performance lithium battery copper foil and electronic circuit copper foil. The company operates four copper foil production bases across Jiangxi, Gansu and Anhui, with a total nameplate capacity of 175 ktpa in 2025. Defu has built a vertically integrated industrial chain covering copper foil production, additives and equipment. Key customers include CATL, ATL, Gotion, EMC, Shengyi Technology, Panasonic and Shennan Circuits. As of 1H26, Lithium battery copper foil and electronic circuit copper foil accounted for 87% and 8% of total gross profit.

Figure 42: Defu: Current shareholding structure



Source: Company data, J.P. Morgan

Table 9: Defu: Management profiles

Name	Title	Experience
Mr. Ma Ke	Chairman	Mr. Ma has served as Chairman of Defu since 2017, after holding senior leadership roles at Defu's predecessor company, including supervisor, general manager, executive director and general manager. He previously served as Chairman of Jiujiang Hongsheng Copper. He holds a bachelor's degree and brings extensive experience in copper-related operations and corporate management.
Mr. Luo Jia	Director and General Manager	Mr. Luo has served as Director and General Manager of Defu since 2017. He holds a PhD in Chemistry from Peking University and previously worked as a postdoctoral researcher and assistant researcher at the Institute of Chemistry, Chinese Academy of Sciences. He also served as General Manager of Beijing Weiming Juli Technology Development and held senior roles at Desi Optoelectronics. He brings strong technical, R&D and management expertise.
Mr. Ma Defu	Director	Mr. Ma has served as a Director of Defu since 2017, after holding multiple senior roles at Defu's predecessor company, including chairman, executive director and general manager, executive director, and supervisor. Earlier in his career, he held leadership positions at Jiujiang Rectifier Factory, Lushan Radio Factory and Jiujiang Electronic Materials Factory. He graduated from Nankai University in Semiconductor Studies, is a senior engineer, and receives the State Council Special Allowance.
Mr. Jin Rongtao	Director and Deputy General Manager	Mr. Jin currently serves as Director and Deputy General Manager of Defu, having joined the company as Deputy General Manager in 2020. He previously held engineering and management roles at Northwest Copper Processing Factory, Tongling Zhongjin Copper Foil, Tongling Nonferrous Tongguan Brass Rod, the China Nonferrous Metals Processing Industry Association and Zhejiang Garden New Energy. He is a professor-level engineer and serves on expert committees for China's nonferrous metals industry, bringing deep technical expertise in copper processing and new materials.

Source: Company report, J.P. Morgan

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 43: Defu: Key Clients

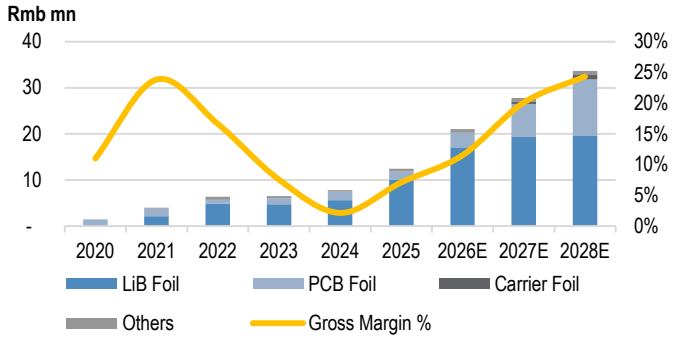
Product	Key Clients
HVLP1-4	TW: Elite Material (EMC), ITEQ CN: Shenyi JP: Panasonic Electronics
LiB copper foil	CN: CATL, Gotion High-Tech, Amperex Technology Limited (ATL), Sunwoda, CALB, Ganfeng Lithium
Carrier-attached copper foil	CN: Shennan circuits TW: Unimicron

Source: Company data, J.P. Morgan.

Financial Analysis

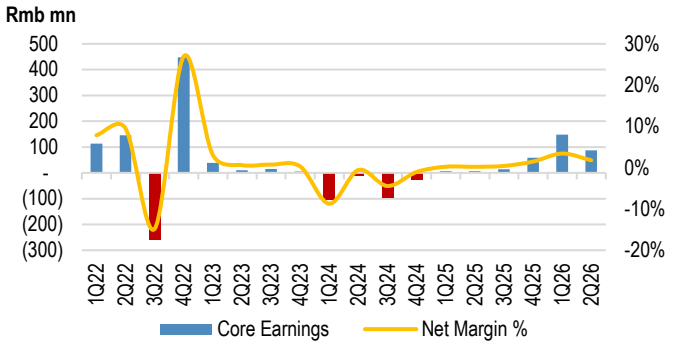
Defu's historical revenue growth was driven mainly by shipment expansion and copper-price pass-through, while profitability was constrained by processing-fee pressure. Copper foil is typically priced on a copper cathode cost plus processing fee basis, with customer settlement and raw-material procurement largely matched to M-1 copper prices, limiting Defu's direct copper-price exposure. From 2020 to 2025, revenue grew at a 54% CAGR from Rmb1.4bn to Rmb12.4bn, supported by a 46% CAGR in copper foil shipments and an 11% CAGR in SHCJ copper prices. Blended ASP rose from c.Rmb65k/t in 2020 to c.Rmb84k/t in 2025, but the increase was largely copper-cost driven; in 2025, ASP comprised c.Rmb69k/t of copper cathode cost and only c.Rmb15k/t of processing fee.

Figure 44: Defu - Revenue Breakdown



Source: Company data, J.P. Morgan estimates

Figure 45: Defu - Core earnings and NPM

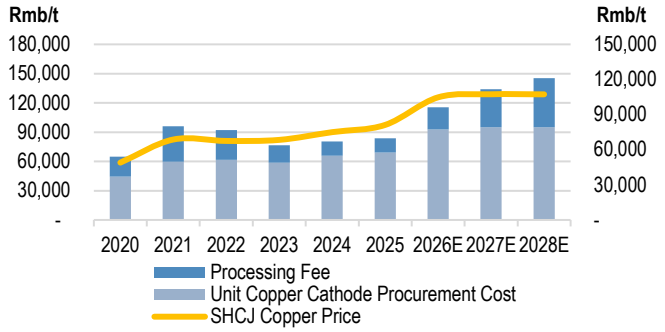


Note: Quarterly core earnings and net margin are estimated numbers
Source: Company data, J.P. Morgan estimates

Profitability was historically more volatile, as industry oversupply and lower processing fees offset part of the volume and copper-price benefit. Defu's blended processing fee declined from c.Rmb21k/t in 2020 to c.Rmb15k/t in 2025, while unit GP was broadly flat at c.Rmb5.9k/t in 2025 versus c.Rmb7.0k/t in 2020. Group gross margin fell from 9.7% in 2020 to 6.8% in 2025, and NPAT to common shareholders remained low at Rmb113mn in 2025 after a loss in 2024. Looking ahead, as Defu shifts toward higher-end PCB copper foil, carrier copper foil and recovering LiB copper foil, we forecast group gross margin to improve to 11%/20%/25% and NPAT to common shareholders to rise to Rmb737mn/Rmb2.44bn/Rmb3.82bn over the same period.

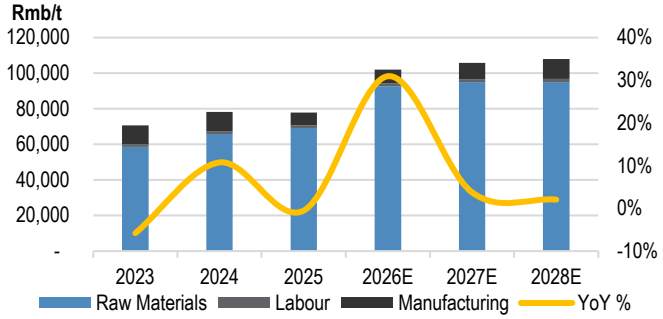
Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 46: Defu - ASP breakdown and copper cost



Source: Company data, J.P. Morgan estimates

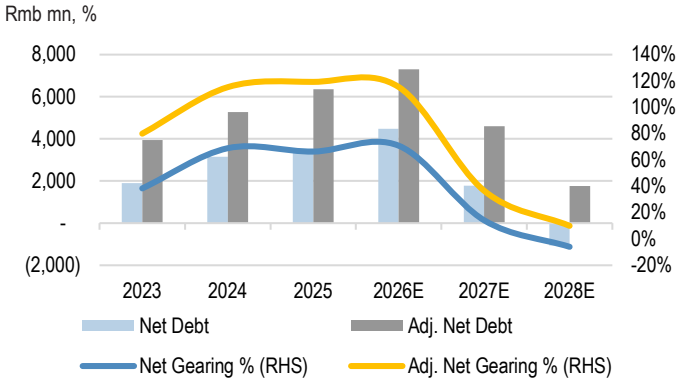
Figure 47: Defu - Unit COGS breakdown



Source: Company data, J.P. Morgan estimates

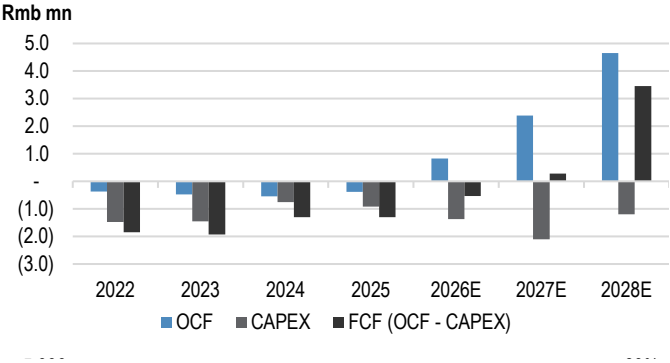
Defu remains in an investment-heavy phase, but its balance sheet should start improving from 2027E as earnings and cash generation catch up with capex. FCF was negative throughout 2020–25, with annual outflows of c.Rmb0.5bn–1.9bn driven by capacity expansion and working-capital needs. We expect FCF to turn positive by end-2026E and continue improving in 2027–28E, supported by higher utilization, margin expansion, and better working capital as the mix of Japanese and Taiwanese customers increases. This customer mix shift should improve cash conversion, as domestic customers typically use 60–90 day bill settlement while Japanese and Taiwanese customers settle monthly. Assuming the Rmb2.8bn placement is completed in 1Q27E, we expect Defu to deleverage steadily in 2027E-2028E.

Figure 48: Defu - Net Gearing Trend



Note: Adjusted net debt and adjusted net gearing exclude restricted cash.
Source: Company data, J.P. Morgan estimates.

Figure 49: Defu - Free cashflow trend

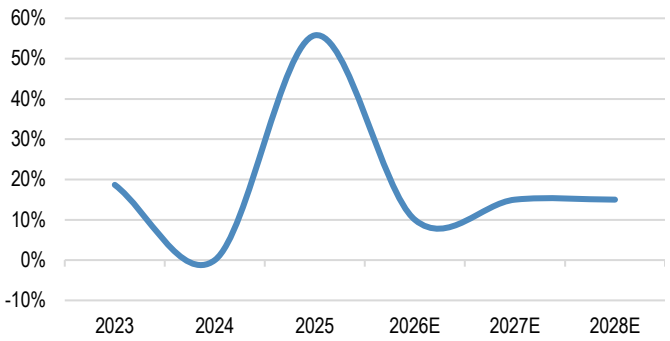


Source: Company data, Bloomberg Finance L.P, J.P. Morgan estimates.

Defu's shareholder return has been modest, but DPS should rise with earnings growth. The company paid its first dividend in 2023 with a 19% payout, suspended dividends in 2024 due to losses, and resumed in 2025 with a 56% payout. We expect payout to normalize at c.10%-15% over 2026–28E as Defu balances dividends with capex and deleveraging. On our estimates, DPS increases to Rmb0.12/Rmb0.56/ Rmb0.87 in 2026E/2027E/2028E, supported by NPAT recovery. Capital returns are not the core investment driver, but improving FCF and lower leverage from 2027E should increase flexibility over time.

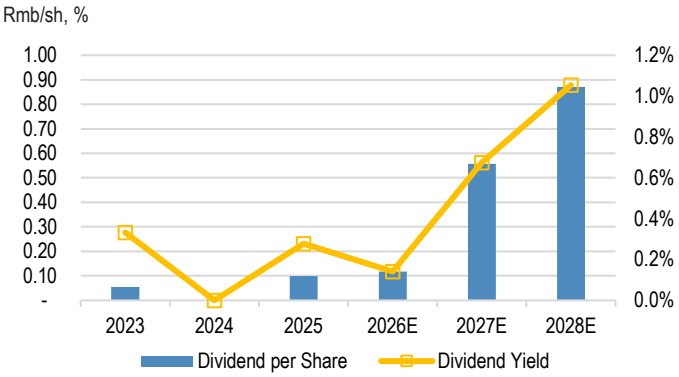
Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Figure 50: Defu: Dividend payout history and forecasts



Source: Company data, J.P. Morgan estimates

Figure 51: Defu: Dividend per share and yields



Source: Company data, J.P. Morgan estimates

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Table 10: Defu - Income statement

Rmb mn unless specified	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Income Statement									
Revenue	**1,427**	**3,986**	**6,381**	**6,531**	**7,805**	**12,437**	**21,024**	**27,802**	**33,632**
YoY %	*n.a.*	*179%*	*60%*	*2%*	*20%*	*59%*	*69%*	*32%*	*21%*
COGS	1,282	3,062	5,319	6,037	7,638	11,552	18,601	22,216	25,448
Business Tax & Surchages	6	16	23	30	31	44	63	97	135
Gross Profit	**139**	**908**	**1,039**	**464**	**136**	**841**	**2,360**	**5,489**	**8,049**
SG&A	(92)	(148)	(208)	(266)	(338)	(439)	(773)	(1,064)	(1,338)
Other Revenue/ (Expense)	26	14	67	151	100	123	-	-	-
EBIT	**72**	**774**	**898**	**349**	**(102)**	**525**	**1,586**	**4,424**	**6,711**
YoY %	*n.a.*	*957%*	*14%*	*-64%*	*-146%*	*-436%*	*223%*	*179%*	*52%*
Depreciation & Amortization	71	98	168	258	339	404	419	490	608
EBITDA	**143**	**872**	**1,065**	**607**	**238**	**929**	**2,006**	**4,914**	**7,319**
Non-operating Revenue/ (Expense)	0	(7)	(21)	(30)	(45)	(34)	-	-	-
Net Interest Expense	(69)	(114)	(160)	(193)	(207)	(291)	(345)	(286)	(245)
Profit Before Tax	**3**	**653**	**717**	**126**	**(353)**	**200**	**1,241**	**4,139**	**6,466**
Income Tax	2	(94)	(79)	(13)	16	46	(180)	(621)	(970)
Minority Interests	13	(93)	(135)	20	92	(133)	(324)	(1,073)	(1,677)
NPAT to Shareholders	**18**	**466**	**503**	**133**	**(245)**	**113**	**737**	**2,444**	**3,819**
YoY %	*n.a.*	*2447%*	*8%*	*-74%*	*-285%*	*-146%*	*555%*	*232%*	*56%*
Fully-diluted EPS (Rmb)	0.08	1.27	1.32	0.33	(0.39)	0.18	1.17	3.71	5.80
Total DPS (Rmb)	-	-	-	0.06	-	0.10	0.12	0.56	0.87
Payout Ratio	0%	0%	0%	19%	0%	56%	10%	15%	15%
Margins & Ratios									
Gross Margin	10%	23%	16%	7%	2%	7%	11%	20%	24%
EBITDA Margin	10%	22%	17%	9%	3%	7%	10%	18%	22%
EBIT Margin	5%	19%	14%	5%	-1%	4%	8%	16%	20%
Net Margin	1%	12%	8%	2%	-3%	1%	4%	9%	11%
ROE %	3%	37%	23%	4%	-6%	3%	17%	33%	33%
Key Operational Data									
Copper Foil Volume (t)	21,082	39,391	61,834	79,108	92,701	140,892	176,060	201,849	225,802
YoY %		*87%*	*57%*	*28%*	*17%*	*52%*	*25%*	*15%*	*12%*
Copper Foil ASP (Rmb/g)	64,969	96,035	92,115	76,640	80,406	83,799	115,480	133,965	145,231
YoY %		*48%*	*-4%*	*-17%*	*5%*	*4%*	*38%*	*16%*	*8%*
Copper Foil unit GP (Rmb/g)	6,983	23,335	17,080	5,950	2,109	5,900	13,429	27,355	35,927
YoY %		*234%*	*-27%*	*-65%*	*-65%*	*180%*	*128%*	*104%*	*31%*

Source: Company data, J.P. Morgan estimates

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

Table 11: Defu - Balance sheet

Rmb mn unless specified	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Balanace Sheet									
Current Assets									
Cash & Cash Equivalent	114	875	1,493	3,539	2,832	5,058	6,681	4,903	8,252
Account Receivables	905	1,397	2,036	2,550	3,619	4,831	5,904	7,588	8,929
Inventory	329	638	1,152	1,740	1,538	1,872	2,982	3,680	4,360
Other Current Assets	62	222	185	222	363	540	680	725	761
Total	**1,410**	**3,131**	**4,866**	**8,051**	**8,352**	**12,302**	**16,247**	**16,896**	**22,302**
Non-current Assets									
PP&E	1,210	1,754	3,246	4,292	5,599	5,961	6,908	8,519	9,111
Other Non-current Assets	374	916	1,769	1,730	929	1,281	1,281	1,281	1,281
Total	**1,584**	**2,670**	**5,014**	**6,022**	**6,527**	**7,242**	**8,189**	**9,800**	**10,392**
Total Assets	**2,994**	**5,801**	**9,880**	**14,074**	**14,880**	**19,545**	**24,436**	**26,696**	**32,694**
Current Liabilities									
Account Payables	913	1,049	2,214	1,760	1,935	2,693	4,001	4,486	4,841
Short-term Borrowings	232	700	2,236	4,249	5,131	7,371	9,939	5,464	5,973
Other Current Liabilities	531	725	949	1,547	1,653	1,933	1,950	1,956	1,960
Total	**1,675**	**2,474**	**5,399**	**7,555**	**8,719**	**11,997**	**15,890**	**11,906**	**12,774**
Non-Current Liabilities									
Long-term Borrowings	395	1,003	1,159	1,186	855	1,221	1,221	1,221	1,221
Other Non-current Liabilities	113	117	231	394	731	1,002	1,002	1,002	1,002
Total	**508**	**1,120**	**1,390**	**1,580**	**1,585**	**2,223**	**2,223**	**2,223**	**2,223**
Shareholders' Equity									
Minority Interests	157	316	696	676	584	1,227	1,551	2,624	4,301
Shareholders' Equity	497	1,575	1,699	3,586	3,407	2,871	3,221	7,319	9,094
Total	**654**	**1,891**	**2,395**	**4,262**	**3,991**	**4,098**	**4,772**	**9,943**	**13,395**
Total Liab. & Shareholder's Equity	**2,994**	**5,801**	**9,880**	**14,074**	**14,880**	**19,545**	**24,436**	**26,696**	**32,694**
Ratios									
Book Value per Share (Rmb)	2.86	5.15	6.28	10.60	6.35	6.50	7.57	15.10	20.35
Net Gearing %	63%	38%	62%	38%	69%	66%	71%	14%	-6%

Source: Company data, J.P. Morgan estimates

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Table 12: Defu - Cashflow statement

Rmb mn unless specified	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash Flow Statement									
Operating Cash Flow									
Net Profit	18	466	503	133	(245)	113	737	2,444	3,819
Depreciation & Amortization	71	98	168	258	339	404	419	490	608
Change in Net Working Capital	353	668	(10)	1,554	730	846	886	1,937	1,696
Other Operating Cash Flow	(821)	(1,542)	(1,033)	(2,421)	(1,374)	(1,744)	(1,214)	(2,487)	(1,473)
Total	**(379)**	**(310)**	**(372)**	**(477)**	**(550)**	**(381)**	**829**	**2,383**	**4,650**
Investing Cash Flow									
Capex	(132)	(900)	(1,477)	(1,457)	(750)	(916)	(1,366)	(2,100)	(1,200)
Other Investing Cash Flow	31	(121)	132	(23)	(158)	157	-	-	-
Total	**(102)**	**(1,021)**	**(1,345)**	**(1,479)**	**(908)**	**(759)**	**(1,366)**	**(2,100)**	**(1,200)**
Financing Cash Flow									
Net Debt Issue/ (Repayment)	(2)	1,093	1,687	2,042	498	1,869	2,569	(4,475)	508
Net Share Issue/ (Buyback)	266	835	245	1,790	-	500	-	2,800	-
Dividend Payment	29	63	96	44	96	169	(63)	(74)	(367)
Other Financing Cash Flow	170	30	(615)	(898)	89	137	(346)	(312)	(243)
Total	**462**	**2,020**	**1,413**	**2,979**	**683**	**2,674**	**2,160**	**(2,061)**	**(101)**
Net Cash Flow	**(19)**	**690**	**(304)**	**1,023**	**(775)**	**1,534**	**1,623**	**(1,778)**	**3,349**
Free Cash Flow	**(512)**	**(1,209)**	**(1,849)**	**(1,933)**	**(1,300)**	**(1,298)**	**(538)**	**283**	**3,450**

Source: Company data, J.P. Morgan estimates.

Avery Chan AC
(852) 2800-8659
avery.chan@jpmorgan.com

Defu - A *(Overweight; Price Target: Rmb108.00)*

Investment Thesis

We initiate coverage of Defu Technology with an OW rating. Defu is a leading Chinese copper foil producer with a large LiB copper foil base and increasing exposure to higher-margin electronic circuit copper foil. After a meaningful share-price correction, we see an attractive entry point ahead of an earnings growth cycle. We expect growth to be driven by PCB copper foil mix upgrades and capacity expansion, carrier copper foil commercialization, and a recovery in LiB copper foil profitability. While the market remains focused on the pace of earnings realization, we think key leading indicators — downstream certification, volume ramp and product mix improvement — are moving in the right direction, supporting potential earnings acceleration and valuation re-rating.

Valuation

We derive our Dec-27 PT of Rmb108/share using a SOTP framework. We value PCB and carrier copper foil at 20x/25x FY28E P/E, lithium battery copper foil at 15x FY28E P/E, and other businesses at 10x P/E, reflecting differing growth prospects and profitability profiles across the portfolio. Our target price implies 29x/19x FY27E/FY28E P/E, between pure PCB copper foil and battery copper foil multiples, reflecting Defu's improving mix and earnings growth.

Table 14: Defu - SOTP Valuation

Breakdown	Implied Value (Rmb/sh)	% of target price	Valuation methodology
PCB Copper Foil	67.5	62%	20x FY28E segment P/E
Carrier Copper Foil	11.9	11%	25x FY28E segment P/E
Lithium Battery Copper Foil	28.5	26%	15x FY28E segment P/E
Other Business	0.5	0%	10x FY28E segment P/E
Target price	**108**	**100%**	

Source: Company data, J.P. Morgan estimates.

Risks to Rating and Price Target

Industry-wide risks include slower-than-expected AI server deployment, weaker demand for next-generation high-speed PCBs, faster-than-expected high-end copper foil capacity additions, and weaker LiB copper foil processing-fee recovery. Company-specific risks include: i) unexpected yield or qualification challenges during Defu's HVLP4 ramp, ii) delays in carrier copper foil commercialization, iii) slower utilization recovery in LiB copper foil, and iv) higher-than-expected balance-sheet pressure during the capacity expansion phase.

Avery Chan [AC]
(852) 2800-8659
avery.chan@jpmorgan.com

J.P.Morgan

Defu - A: Summary of Financials

Income Statement	FY24A	FY25A	FY26E	FY27E	FY28E
Revenue	**7,805**	**12,437**	**21,024**	**27,802**	**33,632**
COGS	(7,638)	(11,552)	(18,601)	(22,216)	(25,448)
Gross profit	**136**	**841**	**2,360**	**5,489**	**8,049**
SG&A	(338)	(439)	(773)	(1,064)	(1,338)
Adj. EBITDA	**193**	**895**	**2,006**	**4,914**	**7,319**
D&A	(339)	(404)	(419)	(490)	(608)
Adj. EBIT	**(146)**	**491**	**1,586**	**4,424**	**6,711**
Net Interest	(148)	(205)	(263)	(204)	(163)
Adj. PBT	**(353)**	**200**	**1,241**	**4,139**	**6,466**
Tax	16	46	(180)	(621)	(970)
Minority Interest	92	(133)	(324)	(1,073)	(1,677)
Adj. Net Income	**(304)**	**(32)**	**737**	**2,444**	**3,819**
Reported EPS	(0.39)	0.18	1.17	3.71	5.80
Adj. EPS	**(0.48)**	**(0.05)**	**1.17**	**3.71**	**5.80**
DPS	**0.00**	**0.10**	**0.12**	**0.56**	**0.87**
Payout ratio	0.0%	55.8%	10.0%	15.0%	15.0%
Shares outstanding	629	630	630	658	658

Balance Sheet	FY24A	FY25A	FY26E	FY27E	FY28E
Cash and cash equivalents	2,832	5,058	6,681	4,903	8,252
Accounts receivable	3,619	4,831	5,904	7,588	8,929
Inventories	1,538	1,872	2,982	3,680	4,360
Other current assets	5,520	7,244	9,566	11,993	14,049
Current assets	**8,352**	**12,302**	**16,247**	**16,896**	**22,302**
PP&E	5,599	5,961	6,908	8,519	9,111
LT investments	0	0	0	0	0
Other non current assets	929	1,281	1,281	1,281	1,281
Total assets	**14,880**	**19,545**	**24,436**	**26,696**	**32,694**
Short term borrowings	5,131	7,371	9,939	5,464	5,973
Payables	1,935	2,693	4,001	4,486	4,841
Other short term liabilities	1,653	1,933	1,950	1,956	1,960
Current liabilities	**8,719**	**11,997**	**15,890**	**11,906**	**12,774**
Long-term debt	855	1,221	1,221	1,221	1,221
Other long term liabilities	1,585	2,223	2,223	2,223	2,223
Total liabilities	**10,304**	**14,220**	**18,113**	**14,129**	**14,997**
Shareholders' equity	3,991	4,098	4,772	9,943	13,395
Minority interests	584	1,227	1,551	2,624	4,301
Total liabilities & equity	**14,880**	**19,545**	**24,436**	**26,696**	**32,694**
BVPS	**6.35**	**6.50**	**7.57**	**15.10**	**20.35**
y/y Growth	*(40.1%)*	*2.4%*	*16.5%*	*99.5%*	*34.7%*
Net debt/(cash)	3,154	3,533	4,479	1,782	(1,059)

Cash Flow Statement	FY24A	FY25A	FY26E	FY27E	FY28E
Cash flow from operating activities	**(550)**	**(381)**	**829**	**2,383**	**4,650**
o/w Depreciation & amortization	339	404	419	490	608
o/w Changes in working capital	(730)	(846)	(997)	(1,937)	(1,696)
Cash flow from investing activities	**(908)**	**(759)**	**(1,366)**	**(2,100)**	**(1,200)**
o/w Capital expenditure	(750)	(916)	(1,366)	(2,100)	(1,200)
as % of sales	*9.6%*	*7.4%*	*6.5%*	*7.6%*	*3.6%*
Cash flow from financing activities	**683**	**2,674**	**2,160**	**(2,061)**	**(101)**
o/w Dividends paid	(129)	(100)	(408)	(386)	(610)
o/w Shares issued/(repurchased)	0	500	0	2,800	0
o/w Net debt issued/(repaid)	498	1,869	2,569	(4,475)	508
Net change in cash	**(775)**	**1,528**	**1,623**	**(1,778)**	**3,349**
Adj. Free cash flow to firm	**(1,300)**	**(1,298)**	**(538)**	**283**	**3,450**
y/y Growth	*(32.8%)*	*(0.1%)*	*(58.6%)*	*(152.7%)*	*1118.7%*

Ratio Analysis	FY24A	FY25A	FY26E	FY27E	FY28E
Gross margin	1.7%	6.8%	11.2%	19.7%	23.9%
EBITDA margin	2.5%	7.2%	9.5%	17.7%	21.8%
EBIT margin	(1.9%)	3.9%	7.5%	15.9%	20.0%
Net profit margin	(3.9%)	(0.3%)	3.5%	8.8%	11.4%
ROE	(7.4%)	(0.8%)	16.6%	33.2%	32.7%
ROA	(2.1%)	(0.2%)	3.4%	9.6%	12.9%
ROCE	(1.6%)	3.3%	9.5%	23.1%	30.7%
SG&A/Sales	4.3%	3.5%	3.7%	3.8%	4.0%
Net debt/Equity	0.7	0.7	0.7	0.1	NM
Net debt/EBITDA	16.3	3.9	2.2	0.4	NM
Sales/Assets (x)	0.5	0.7	1.0	1.1	1.1
Assets/Equity (x)	3.5	4.3	5.0	3.5	2.5
Interest cover (x)	1.3	4.4	7.6	24.1	44.9
Operating leverage	(747.8%)	(734.6%)	323.1%	554.8%	246.5%
Tax rate	(4.7%)	22.8%	14.5%	15.0%	15.0%
Revenue y/y Growth	19.5%	59.3%	69.0%	32.2%	21.0%
EBITDA y/y Growth	(66.5%)	363.8%	124.2%	145.0%	48.9%
EPS y/y Growth	7235.7%	(89.6%)	(2425.4%)	217.4%	56.2%

Valuation	FY24A	FY25A	FY26E	FY27E	FY28E
P/E (x)	NM	NM	66.8	21.1	13.5
P/BV (x)	12.3	12.0	10.3	5.2	3.8
EV/EBITDA (x)	274.8	60.4	27.6	10.9	7.2
Dividend Yield	0.0%	0.1%	0.1%	0.7%	1.1%

Source: Company reports and J.P. Morgan estimates.
Note: Rmb in millions (except per-share data).Fiscal year ends Dec. o/w - out of which

Companies Discussed in This Report (all prices in this report as of market close on 15 October 2026, unless otherwise indicated)
CATL - H(3750.HK/HK$448.60/OW), Jiangxi Copper - A(600362.SS/Rmb38.90/OW), Jiangxi Copper - H(0358.HK/HK$31.20/OW), LG Energy Solution(373220.KS/W403,500/OW), MMG(1208.HK/HK$7.87/OW), Samsung SDI(006400.KS/W543,000/OW), Zijin Mining - A(601899.SS/Rmb29.38/OW), Zijin Mining – H(2899.HK/HK$31.26/OW)

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Defu - A or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Defu - A or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Defu - A or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Defu - A or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:
J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, https://www.jpmorganmarkets.com.

Coverage Universe: Chan, Avery : Angang Steel - A (000898.SZ), Angang Steel - H (0347.HK), Baosteel – A (600019.SS), CMOC - A (603993.SS), CMOC - H (3993.HK), Chalco - A (601600.SS), Chalco - H (2600.HK), China Hongqiao - H (1378.HK), China Jushi - A (600176.SS), Ganfeng Lithium – A (002460.SZ), Ganfeng Lithium – H (1772.HK), Jiangxi Copper - A (600362.SS), Jiangxi Copper - H (0358.HK), MMG (1208.HK), Shandong Gold - A (600547.SS), Shandong Gold - H (1787.HK), Shenhua - A (601088.SS), Shenhua - H (1088.HK), Tianqi Lithium – A (002466.SZ), Tianqi Lithium – H (9696.HK), Yankuang - A (600188.SS), Yankuang - H (1171.HK), Zijin Gold International - H (2259.HK), Zijin Mining - A (601899.SS), Zijin Mining – H (2899.HK)

J.P. Morgan Equity Research Ratings Distribution, as of October 03, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	35%	12%
IB clients**	83%	81%	72%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	93%	88%

***Please note that the percentages may not add to 100% because of rounding.**
****Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.**
For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at http://www.jpmorganmarkets.com, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, http://www.jpmorganmarkets.com. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Digital assets, which may also be referred to as virtual or crypto assets or crypto currencies (e.g. bitcoin), discussed in this research report are subject to a rapidly changing regulatory landscape. The application of relevant laws in respect of such assets may be unsettled, and the legality of any digital asset investment may be unclear. Prior to making any investment decision regarding any digital asset, investors should seek their own legal advice and be aware of any legal considerations regarding the purchase, holding or selling of such an investment. For relevant regulatory advisories on digital assets, including bitcoin and ethereum, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the

ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is

authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Nigeria: J.P. Morgan Chase Representative Office (Nigeria) Limited is regulated by the Central Bank of Nigeria.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 070/08/2026 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumphini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Any mentions of third-party trademarks, brand names, products, and services are for referential purposes only and any mention thereof is not meant to imply any sponsorship, endorsement, or affiliation.

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised October 03, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 16 Oct 2026 02:09 AM HKT

Disseminated 16 Oct 2026 02:10 AM HKT